CERTIFYING WHAT YOUR GENERATIVE MODEL COVERS: LABELLER-CONDITIONAL, FINITE-SAMPLE MODE-RECALL GUARANTEES

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ABSTRACT

Recall- and coverage-style metrics are the standard tools for asking whether a generative model has silently *dropped modes*, yet they carry no guarantee and entangle fidelity with coverage—a collapsed GAN and a fully-covering flow model can both get nearest-neighbour recall ≈ 0, and in controlled settings recall and FID instead *over*-state coverage. We introduce *certified mode-recall*: given a mode labeller (a class or attribute classifier) and a tolerance γ, a finite-sample, *two-sided* per-mode procedure that certifies each mode **present** ($\rho_j \geq \gamma$), **absent** ($\rho_j < \gamma$, genuine collapse *detected*), or **inconclusive**—so we both lower-bound coverage and detect missing modes, rather than reporting a floor a low value cannot distinguish from under-sampling. It is *labeller-conditional* and otherwise distribution-free (assuming only a bound on the labeller's confusion on the generator), with per-mode thresholds made rigorous by *sample splitting* and a *deconvolution* to true (not predicted) mode frequencies; it certifies presence/absence of labeller-defined subgroups, not within-subgroup diversity. We prove two-sided validity; that the $\gamma\widehat{q}_j^{\mathrm{up}}$ threshold closes the otherwise uncertifiable band $[\gamma, \gamma/q_j]$; γ-presence consistency for the LP variant (the default deconvolved rule stays valid but conservative by a term second order in the labeller's confusion); and a witnessing rate $n \gtrsim \log(M/\delta)/(\rho_j q_j)^2$ (in the leakage-dominated regime; $1/(\rho_j q_j)$ under low leakage) that delimits when rare protected subgroups are certifiable at all. Empirically the certificate is valid (zero false present/absent over seeds) and detects collapse across GANs, flow-matching, and pretrained diffusion (CIFAR DDPM, CelebA-HQ 256), exposing over- and under-statement that precision/recall, density/coverage, and FID all miss.

1 INTRODUCTION

Modern image generators—diffusion and flow-matching models in particular—produce strikingly realistic samples, but realism is not coverage. A model can generate beautiful images while silently omitting entire regions of the data distribution: rare classes, minority subpopulations, or specific attribute combinations. Detecting such *mode dropping* is the job of recall/coverage metrics (Kynkäänniemi et al., 2019; Naeem et al., 2020). These metrics estimate a manifold from samples via k-nearest-neighbour hyperspheres and measure how much of the real manifold falls inside the generated one.

Our starting observation is that this manifold-based estimate is *not a guarantee, and is systematically optimistic*. In a controlled setting where the generator provably covers only a fraction $1-f$ of the data's modes, improved recall and density/coverage report values far above $1-f$ (Table 6): at $f{=}0.5$ they report near-1.0 coverage while half the modes are absent. The cause is geometric: distinct modes overlap in feature space, so the hyperspheres of the modes the generator *does* cover also enclose real points from the modes it does *not*. FID inherits the same blindness, moving only weakly. A practitioner who reads these numbers concludes the model is fine when it is not.

We argue mode coverage should be *certified*, not scored. Given a mode labeller $\widehat{m}$ that partitions samples into M semantic modes (a class or attribute classifier), we ask for a finite-sample lower

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confidence bound $\widehat{R}_{\mathrm{lo}}$ on the mode recall R—the fraction of *labeller* modes the generator covers—such that $\Pr[R \geq \widehat{R}_{\mathrm{lo}}] \geq 1 - \delta$. This is a *labeller-conditional* guarantee: we make no parametric assumption on the data, generator, or feature space, but we do assume a bound on the labeller's confusion (Assumption 1); with a perfect labeller it is fully distribution-free. The semantics matter—we certify coverage of the labeller's partition, so the certificate is exactly as meaningful as the modes the auditor cares to define. To our knowledge no such image-side certificate exists; the guarantees are for discrete/LLM outputs via missing-mass arguments (Noorani et al., 2025). Our sample-complexity analysis (§4.1) is frank about the cost: in the leakage-dominated worst case, certifying a subgroup of mass ρ_j under a labeller of accuracy q_j needs $n \gtrsim 1/(\rho_j q_j)^2$ samples (improving to $1/(\rho_j q_j)$ under low leakage), so *rare* protected subgroups—the very case fairness auditing cares about most—are the most expensive to certify, a limit no distribution-free method escapes.

Contributions.

- **An exposure of the incumbent failure.** We give a controlled benchmark that isolates mode-dropping with *known* ground truth, and show that nearest-neighbour recall/coverage and FID substantially over-state coverage—a safety-relevant failure, not merely a looseness (§3).
- **Certified mode-recall.** A mode-resolved, labeller-conditional, finite-sample lower bound on mode recall. Coverage is decided *per mode* by a count threshold made robust to an imperfect labeller via a confusion-aware, *sample-split* multiple-testing rule with a deconvolution correction for the labeller's confusion (§3).
- **Guarantees.** We prove finite-sample validity (Bernstein-slacked, not merely in expectation), *consistency* for the LP variant (with an explicit second-order gap for the default deconvolved rule), and a *sample-complexity rate*: a general sufficient $n \gtrsim \log(M/\delta)/(\rho_j q_j)^2$, tight (with a matching lower bound) in the leakage-dominated regime—quantifying when rare subgroups are certifiable (§4–4.1).
- **Evidence.** Across the 1000-mode StackedMNIST benchmark (with 12-seed error bars), controlled synthetic subgroup benchmarks, and real generators of *three families*—GANs, flow matching (probability-flow ODE), and *pretrained diffusion* (CIFAR DDPM, CelebA-HQ 256)—the certificate is valid, tight where the budget allows, and flags dropping invisible to precision/recall, density/coverage, and FID (§5).

Confusion-corrected prevalence estimation (quantification / label shift) is a point estimate; our technical increment is to turn it into a certificate through a specific combination that a prevalence CI does not provide: *simultaneous two-sided per-mode decisions* (present/absent/inconclusive), a *finite-sample confusion-aware construction* (empirical-Bernstein deconvolved thresholds rather than a point or normal-approximation interval), the $\gamma\widehat{q}_j^{\mathrm{up}}$ *consistency correction*, a *robust LP* that stays valid under confusion shift, and *generated-side audit calibration*.

2 Related Work

Generative evaluation. Fréchet Inception Distance (Heusel et al., 2017) and the precision/recall family (Sajjadi et al., 2018; Kynkäänniemi et al., 2019) and density/coverage (Naeem et al., 2020) are the standard diversity/fidelity diagnostics. All are point estimates from finite samples with no coverage guarantee; we show the recall/coverage members are optimistically biased under mode dropping.

Quantification and label shift. Our deconvolved leakage estimate is confusion-corrected *prevalence estimation*, the core of quantification learning (González et al., 2017) and label-shift correction (Saerens et al., 2002; Lipton et al., 2018). These return a *point* prevalence with *no* finite-sample validity—a BBSE-style estimate thresholded at γ over-certifies (Table 9). The statistical primitives we use are all standard; the novelty is their *composition into the first finite-sample two-sided mode-coverage certificate for a generative model* (present via an empirical-Bernstein leakage bound, absent via a q-lower bound, sample splitting, Bonferroni), plus the $\gamma\widehat{q}^{\mathrm{up}}$ consistency correction and

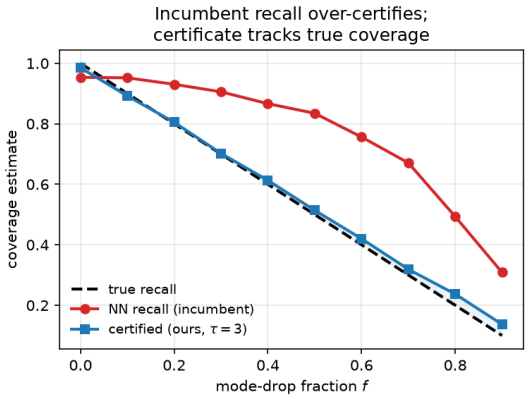


Figure 1: Dropping a fraction f of 1000 StackedMNIST modes: NN recall (red) stays far above the truth even when most modes are gone, while our certificate (blue) tracks the true recall (dashed) from below—a valid, tight floor.

audit-set calibration. When the confusion matrix is well-specified even a properly built adjusted-prevalence CI is a valid, equally tight baseline (Table 9); our increment is the certify/detect framing and validity without matrix inversion or normality (Table 8), not a tighter estimator in the benign regime.

Fairness of generative models. A growing line audits generative models for demographic or subgroup imbalance (Choi et al., 2020; Teo et al., 2023), typically by comparing generated attribute frequencies to reference proportions without a finite-sample guarantee. Our equal-weight certificate provides exactly such a guarantee for subgroup *presence*, and §4.1 shows why rare protected subgroups are the statistically hardest to certify. Appendix F extends this discussion to conformal/risk-control guarantees for generation, mode counting and two-sample tests, and hallucination/typicality detectors.

3 Certified Mode-Recall

We first show why feature-space coverage silently *over-certifies*, then build the certificate that fixes it.

Setup. Let p be the data distribution and $m : \mathcal{X} \to \{1, \dots, M\}$ a mode labeller (e.g. a classifier), inducing mode masses $\pi_j = \Pr_{x\sim p}[m(x) = j]$. A generator q has mode masses $\rho_j = \Pr_{x\sim q}[m(x) = j]$. Define the (mass-weighted) *true mode recall*

$$R = \sum_{j=1}^{M} \pi_j \, \mathbf{1}[\rho_j > 0] \,, \tag{1}$$

the fraction of data mass whose mode the generator can produce. We build a benchmark with M=1000 modes and a generator that provably drops a controlled fraction f of them, so $R = 1 - f$ exactly.

Finding. In a learned feature space (StackedMNIST, classifier penultimate features), nearest-neighbour recall/coverage are badly optimistic—at f=0.5 they report ≈ 0.86–1.0 coverage although $R = 0.5$ (Fig. 1; Appendix D, Table 6)—because covered modes' manifold hyperspheres leak across feature-space overlap and count dropped-mode real points as covered; FID moves weakly. *A fair reading:* recall/coverage use *no* labeller (a harder, label-free problem), whereas our certificate consumes a mode partition; this controlled setup (restricting *real* images to a kept mode set) deliberately isolates coverage from fidelity. The point is not that recall is bad, but that when a mode partition *is* available—whenever an auditor can name the subgroups—one can *certify* coverage rather than read an unguaranteed manifold score. Real-generator evidence is in §5.

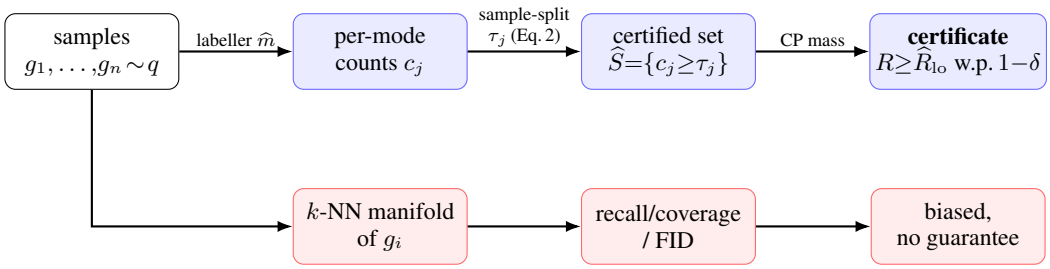


Figure 2: Certified mode-recall (top, blue): label samples into modes, threshold each count against a sample-split confusion-aware level, report a labeller-conditional lower bound. Incumbent manifold metrics (bottom, red) are biased with no guarantee.

The certificate. We now turn the observation above into a guarantee.

Fix a generator q, a mode labeller $\widehat{m}$ (imperfect), a target level δ, and a budget of n generated samples $g_1, \ldots, g_n \sim q$. Let $c_j = \#\{i : \widehat{m}(g_i) = j\}$. Figure 2 summarises the pipeline and Algorithm 1 the procedure.

Idealised certificate (perfect labeller). If $\widehat{m} = m$, then any mode with $c_j \geq 1$ is *demonstrably* covered ($\rho_j > 0$), so the witnessed set $\widehat{S} = \{j : c_j \geq 1\}$ satisfies $\widehat{S} \subseteq S^\star = \{j : \rho_j > 0\}$ deterministically, and $\sum_{j \in \widehat{S}} \widehat{\pi}_j$ is a valid lower bound on R (up to estimating π_j from real data, handled by a Clopper–Pearson step). Rare covered modes may be missed at finite n; this looseness is characterized by the consistency and sample-complexity results of §4.1.

Confusion-aware threshold (imperfect labeller). With label noise a truly uncovered mode j ($\rho_j = 0$) can still receive samples through misclassification, with expected count $\mu_j = n \sum_{t \neq j} \rho_t \, P(\widehat{m}{=}j \mid m{=}t) \leq n\,\varepsilon$, where ε bounds per-mode leakage (estimated from a held-out labelled confusion matrix). We certify a mode only if its count clears a threshold τ that makes false inclusion improbable:

$$\widehat{S}_\tau = \{j : c_j \geq \tau\}, \qquad \tau = \min\Big\{t : \Pr[\mathrm{Bin}(n, \eta) \geq t] \leq \delta / M\Big\}, \tag{2}$$

a Bonferroni control across the M modes ($\mathrm{Bin}(n, \eta)$ throughout; the Poisson$(n\eta)$ tail is the standard rare-event approximation when η is small). The reported certificate is $\widehat{R}_{\text{lo}} = \underline{w}(\widehat{S}_\tau)$, where $\underline{w}$ is a Clopper–Pearson lower bound on $\sum_{j \in \widehat{S}_\tau} \pi_j$ from an independent real calibration sample.

Assumption 1 (Bounded label leakage)**.** *Let $Q(t, j) = \Pr_{x \sim q}[\widehat{m}(x) = j \mid m^\star(x) = t]$ be the labeller confusion* on generated samples. *There is a known η with $Q(t, j) \leq \eta$ for all $t \neq j$. Then for any uncovered mode j ($\rho_j = 0$), a generated sample is labelled j with probability $p_j = \sum_t \rho_t Q(t, j) \leq \eta$.*

We estimate $\widehat{Q}$ on held-out *real* labelled data; since we need confusion on q, we inflate it by a slack that accounts for generator–data distribution shift (discussed below and in Appendix C). The worst-case rate η in (2) is loose when a few mode pairs are highly confusable (e.g. cat/dog): a single large off-diagonal forces a large τ for *all* modes. We therefore use a *per-mode, sample-split* threshold. The leakage into an uncovered mode j is $p_j = \sum_{t \neq j} \rho_t Q(t, j)$, which needs the generator's *true* mode masses ρ_t—not the predicted-label frequencies. We recover them by *deconvolution*: since $\Pr_q[\widehat{m}{=}t] = \sum_s \rho_s Q(s, t) \geq \rho_t \, q_t$ with $q_t = Q(t, t)$, the predicted-count ratio a_t / q_t is an upper estimate of $n\rho_t$. So from the first batch we form

$$\widehat{\mu}_j = \sum_{t \neq j} \frac{a_t}{\widehat{q}_t} \widehat{Q}(t, j) \;\geq\; n\, p_j \quad \text{(in expectation)}, \qquad \tau_j = \min\{t : \Pr[\mathrm{Bin}(n, \widehat{\mu}_j / n) \geq t] \leq \delta / M\}, \tag{3}$$

and test c_j from the *second* batch: certify j iff $c_j \geq \tau_j$. Because τ_j depends only on the first batch it is independent of c_j, so the Binomial-tail bound applies to the estimated threshold (Thm. 1, Appendix A). The earlier draft used a_t in place of $a_t / \widehat{q}_t$, which targets predicted rather than true

mode frequencies; the deconvolution corrects the estimand. Numerically the correction is small (for our labellers $q_t \approx 0.9$) and the certificate is empirically valid in $120/120$ seed-trials (§5). The point estimate only guarantees $\mathbb{E}[\widehat{\mu}_j] \geq np_j$; we upgrade it to a genuine high-probability upper bound $\widehat{\mu}_j^{\uparrow}$ by adding an *empirical-Bernstein* slack (Maurer–Pontil), so $\Pr[\widehat{\mu}_j^{\uparrow} \geq np_j] \geq 1 - \delta/3M$ and Theorem 1 holds *finite-sample*; at our budgets the slack is negligible (certified present 0.49 either way on the M=32 benchmark). The worst-case η rule (2) is a conservative fallback, and a linear program over a joint (Q, ρ) confidence set gives an *exact* finite-sample misclassification-adjusted bound that stays valid under labeller shift while remaining tighter than η. Exact formulas (Eq. 7), constants, and the LP are in Appendix C (Table 8).

Definition 1 (Absolute and relative γ-coverage)**.** *For a tolerance $\gamma > 0$, mode j is* (γ-)covered *if $\rho_j \geq \gamma$, and the* absolute γ*-coverage is $R_\gamma = \sum_j \pi_j \mathbf{1}[\rho_j \geq \gamma]$. For fairness auditing we also define the* relative *target: mode j is c-*represented *if $\rho_j \geq c\,\pi_j$ (generator mass at least a fraction c of the data mass), giving $R_c^{\mathrm{rel}} = \sum_j \pi_j \mathbf{1}[\rho_j \geq c\pi_j]$. We certify these rather than support-level $R = \sum_j \pi_j \mathbf{1}[\rho_j > 0]$, which is not finitely verifiable and, for continuous generators, trivially near* 1.

Absolute and relative answer different questions and neither dominates (a rare subgroup with $\pi_j < \gamma$ can never pass absolute γ; a compressed common subgroup still can). Our procedure certifies both with no change—the two-sided machinery takes a per-mode level, either the constant γ or c times a Clopper–Pearson bound on π_j, with the present/absent sides using *opposite* bound directions for validity (Appendix C). We report absolute γ as the default; §5 and Appendix D include the relative variant.

Two-sided γ-coverage: present, absent, inconclusive. A lower bound alone certifies presence; it cannot *detect* a missing mode, since a small certified value could equally reflect an insufficient budget. We therefore test each mode *two-sidedly* at tolerance γ (Def. 1), splitting the error δ across the two directions and the M modes:

- **present** ($\rho_j \geq \gamma$, covered): $c_j \geq \tau_j^+$ with τ_j^+ the upper $\frac{\delta}{3M}$-tail of $\mathrm{Bin}(n, \gamma\,\widehat{q}_j^{\,\mathrm{up}} + \widehat{\mu}_j/n)$—rejecting $\rho_j < \gamma$ (the factor $\widehat{q}_j^{\,\mathrm{up}}$, not 1, is what makes this consistent for γ-presence rather than for the shifted target $\rho_j > \gamma/q_j$);
- **absent** ($\rho_j < \gamma$, *collapse detected*): $c_j \leq \tau_j^-$ with τ_j^- the lower $\frac{\delta}{3M}$-tail of $\mathrm{Bin}(n, \gamma\,\widehat{q}_j)$—rejecting $\rho_j \geq \gamma$;
- **inconclusive** otherwise—the budget cannot separate the two at this γ.

This unifies certification and detection: present modes give the coverage lower bound, absent modes are provably dropped, and inconclusive is reported honestly. *Three estimands should not be conflated*: witnessed/support coverage ($c_j \geq 1$), γ*-presence* ($\rho_j \geq \gamma$, the present side), and γ*-absence* ($\rho_j < \gamma$, the absent side); this paper certifies the latter two, which are the operationally meaningful ones. Both sides are jointly valid with probability $\geq 1 - \delta$ (Theorem 2): the present side rejects $\rho_j < \gamma$ using the *upper* leakage $\widehat{\mu}_j^{\uparrow}$ and the *upper* confusion bound $\widehat{q}_j^{\,\mathrm{up}}$, and the absent side rejects $\rho_j \geq \gamma$ using a *lower* bound $\widehat{q}_j^{\,\mathrm{lo}}$ on the correct-label rate—the two sides are analogous but *not* identical (one needs a leakage upper bound, the other a q_j lower bound), and the error budget is split across present/absent, the M modes, and confusion estimation. §5 confirms zero false present/absent over 20 seeds.

Calibrating the shift margin. Assumption 1 bounds confusion *under* q; we set the margin η from a small trusted audit set of generated samples (human or oracle labels), which calibrates the $O(K^2)$ confusion once and amortises into a certificate on a large unlabelled batch—see Appendix C for the procedure, why a label-free TV surrogate is *not* valid, and the rare-mode calibration limit. Absent trusted labels the guarantee is honestly conditional on η, with the worst-case η rule as the safe fallback.

4 VALIDITY

Let $S^\star = \{j : \rho_j > 0\}$ be the truly covered set and $R^{\mathrm{u}} = |S^\star|/M$ the unweighted true recall.

Algorithm 1 Certified mode-recall lower bound

Require: generator q, labeller $\widehat{m}$, held-out labelled set for confusion $\widehat{Q}$ (with CP slack, evaluated on samples from q; §3), budgets n, n', level δ, M modes

draw two *independent* batches $A, B \sim q$ of size n each {sample splitting}

$a_j \leftarrow \#\{i \in A : \widehat{m}(g_i) = j\}$; $c_j \leftarrow \#\{i \in B : \widehat{m}(g_i) = j\}$

for each mode $j = 1, \dots, M$ **do**

  $\widehat{\mu}_j^{\uparrow} \leftarrow$ deconvolved leakage on A with empirical-Bernstein slack, Eq. (7) {genuine high-prob upper bound on np_j}

  $\tau_j \leftarrow \min\{t : \Pr[\mathrm{Bin}(n, \gamma\,\widehat{q}_j^{\mathrm{up}} + \widehat{\mu}_j^{\uparrow}/n) \geq t] \leq \delta/3M\}$ (use $\gamma = 0^+$ for support-only witnessing) {present side, γ-coverage}

end for

$\widehat{S} \leftarrow \{j : c_j \geq \tau_j\}$ {test on B; $\tau_j \perp c_j$, so $\widehat{S} \subseteq \{\rho_j \geq \gamma\}$ w.p. $\geq 1 - \delta$}

draw $x_1, \dots, x_{n'} \sim p$; $b \leftarrow \#\{i : m^\star(x_i) \in \widehat{S}\}$

$\widehat{R}_{\mathrm{lo}} \leftarrow$ CLOPPERPEARSONLOWER(b, n', δ) {mass-weighted; use $|\widehat{S}|/M$ for equal-weight}

return $\widehat{R}_{\mathrm{lo}}$ such that $\Pr[R \geq \widehat{R}_{\mathrm{lo}}] \geq 1 - 2\delta$

Theorem 1 (Validity)**.** *Under Assumption 1 (off-diagonal label confusion at most η, so that for every uncovered mode j each generated sample is labelled j with probability at most η) and τ chosen as in (2) with $n\varepsilon$ replaced by $n\eta$, the certified set satisfies $\Pr[\widehat{S}_\tau \subseteq S^\star] \geq 1 - \delta$. Consequently the reported unweighted certificate $\widehat{R}^{\mathrm{u}} = |\widehat{S}_\tau|/M$ is a valid lower bound, $\Pr[\widehat{R}^{\mathrm{u}} \leq R^{\mathrm{u}}] \geq 1 - \delta$. The mass-weighted certificate $\widehat{R}_{\mathrm{lo}}$ is valid at level $1 - 2\delta$ (Appendix A).*

The δ-budget in one place. To avoid the reader tracking $1-\delta$ versus $1-2\delta$ across statements: the *unweighted* certificate $\widehat{R}^{\mathrm{u}}$ holds at $1-\delta$, split three ways as $\delta_{\mathrm{leak}} + \delta_{\mathrm{conf}} + \delta_{\mathrm{test}}$ (leakage concentration, confusion CP bounds, count test); the *mass-weighted* bound spends one extra δ on the real-calibration CP step, hence $1-2\delta$; the *two-sided* certificate splits $\delta = \delta_{\mathrm{p}} + \delta_{\mathrm{a}} + \delta_{\mathrm{c}}$. In every case $\delta = 0.05$ is split evenly into three slices of $\delta/3$, and statements, formulas and code use that same split. No reported guarantee mixes them.

We now state the full two-sided guarantee that the abstract advertises. It is *not* perfectly symmetric: the present side needs an *upper* bound on leakage, the absent side a *lower* bound on the correct-label rate q_j.

Theorem 2 (Two-sided γ-certificate)**.** *Fix γ, δ and split δ as $\delta = \delta_{\mathrm{p}} + \delta_{\mathrm{a}} + \delta_{\mathrm{c}}$ across the present side, absent side, and confusion estimation. Use two independent generated batches A (size n, for thresholds) and B (size n, tested), a held-out labelled set for the confusion CP bounds (charged δ_{c}), and let $\widehat{\mu}_j^{\uparrow}$ be the empirical-Bernstein upper leakage of (7) and $\widehat{q}_j^{\mathrm{lo}}, \widehat{q}_j^{\mathrm{up}}$ CP lower/upper bounds on q_j. Define*

$$\textit{present: } c_j \geq \tau_j^{+} = \min\{t : \Pr[\mathrm{Bin}(n, \gamma\,\widehat{q}_j^{\mathrm{up}} + \widehat{\mu}_j^{\uparrow}/n) \geq t] \leq \delta_{\mathrm{p}}/M\},$$
$$\textit{absent: } c_j \leq \tau_j^{-} = \max\{t : \Pr[\mathrm{Bin}(n, \gamma\,\widehat{q}_j^{\mathrm{lo}}) \leq t] \leq \delta_{\mathrm{a}}/M\}.$$

Then with probability $\geq 1 - \delta$ (jointly over all M modes and both sides): every present-certified mode has $\rho_j \geq \gamma$, and every absent-certified mode has $\rho_j < \gamma$. Hence $\widehat{R}_\gamma^{\mathrm{present}}$ is a valid lower bound on the γ-coverage R_γ, and absent certificates are valid collapse detections.

Validity is one-sided by design. *Tightness*—how many covered modes a finite budget can miss—is governed by consistency and the sample-complexity rate below (§4.1); a Good–Turing estimate bounds the generator's own unseen mass but *not* the recall gap (Proposition 1, Appendix A.2).

4.1 Consistency and sample complexity

Validity alone permits a trivial (always-zero) certificate. We now show the certificate is *consistent* and quantify the budget needed to certify a given mode. Write $q_j = Q(j, j)$ for the labeller's correct-label rate on mode j, ρ_j for its generator mass, and $\mu_j = \sum_{t \neq j} \rho_t Q(t, j)$ for the leakage into j.

Three distinct estimands. We separate (E1) support ($\rho_j > 0$), (E2) γ-presence, (E3) γ-absence: Theorem 3 and the rate (4) concern *witnessing* (E1), the two-sided procedure targets (E2)/(E3) (Appendix A.4).

We are precise about what this buys: the deconvolved leakage estimate is itself an upper bound whose limit exceeds the true leakage by a term second order in the confusion and non-vanishing in n, so Algorithm 1 is valid but consistent only up to that term (Prop. 2); full γ-presence consistency belongs to the LP variant (Cor. 1, App. A.4).

Theorem 3 (Consistency)**.** *Fix the generator, labeller and δ, and use the per-mode threshold τ_j with the population leakage μ_j. Then as $n \to \infty$, $\widehat{S}_\tau \to \{j : \rho_j q_j > 0\}$ almost surely. In particular, if $q_j > 0$ for every covered mode (the labeller labels each mode correctly with nonzero probability), then $\widehat{S}_\tau \to S^\star$ and $\widehat{R}^{\mathrm{u}} \to R^{\mathrm{u}}$, $\widehat{R}_{\mathrm{lo}} \to R$ a.s.: the certificate is asymptotically tight while remaining valid at every finite n.*

Stated for the *population* leakage μ_j; the implemented rule inflates it by Δ_j (Prop. 2), so it witnesses $\{j : \rho_j q_j > \Delta_j\}$.

Theorem 4 (Certifiability rate)**.** *There is a universal constant C such that a covered mode j is certified ($j \in \widehat{S}_\tau$) with probability at least $1 - \delta'$ whenever the sample budget satisfies*

$$n \;\geq\; \frac{C\,\log(M/\delta) + C\,\log(1/\delta')}{(\rho_j q_j)^2}\,\big(\mu_j + \rho_j q_j\big). \tag{4}$$

This is a general sufficient *rate. Its behaviour has two regimes, which we state separately to avoid over-claiming. In the* leakage-dominated *regime ($\mu_j \gtrsim \rho_j q_j$)—the hard case for confusable labellers—the effective rate is $n \gtrsim \mu_j \log(M/\delta)/(\rho_j q_j)^2$, i.e. inverse-square in $\rho_j q_j$, and this is tight: no test separates j from leakage when $\rho_j q_j \lesssim \sqrt{\mu_j/n}$ (matching lower bound, Appendix A.4). In the* low-leakage *regime ($\mu_j \to 0$, e.g. a near-perfect labeller) the mode is witnessed at the milder $n \gtrsim \log(M/\delta)/(\rho_j q_j)$ rate; the inverse-square form is then only an upper bound,* not *matching. The simplified $1/(\rho_j q_j)^2$ headline should be read as the leakage-dominated worst case.*

5 Experiments

We validate: **(V)**alidity—neither side over-claims (present never exceeds true coverage, absent never mislabels a covered mode), at rate $\geq 1 - \delta$; **(T)**ightness—the present bound tracks true recall; **(D)**etection—the *absent* certificate provably flags dropped modes, which incumbent metrics (that only report a single unguaranteed score) cannot.

What "true coverage" means. We report the operational γ-*coverage* $R_\gamma = \sum_j \pi_j \mathbf{1}[\rho_j \geq \gamma]$ (support $\rho_j > 0$ is not finitely verifiable); the present side certifies exactly this ($c_j \geq \tau_j^+ \Rightarrow \rho_j \geq \gamma$ at $1 - \delta$), and the "true coverage" baseline is R_γ witnessed in a large pool (leakage-corrected to be a fair oracle; Appendix D). Validity is established where the modes are known *independently* of the labeller (the synthetic and StackedMNIST benchmarks, where π_j uses *true* labels); the natural-image truncation studies are same-labeller consistency checks, and §E adds an independent human audit. Real-generator validity is *conditional on Assumption 1* on q—labeller-conditional, not unconditional.

Significance (StackedMNIST, 1000 **modes,** 12 **seeds).** Figure 7 reports the certified equal-weight bound with per-mode, *sample-split* thresholds, aggregated over 12 seeds at each of 10 drop levels (120 trials) with $\pm$std error bars, and the per-f *validity rate*—the fraction of seeds in which the certificate did not exceed the true recall—with Wilson 95% intervals (Table 1). The certificate is a valid lower bound that tracks the true recall from below across the whole range and drops monotonically with f, whereas nearest-neighbour recall stays far above the truth and only "catches up" cosmetically as fidelity collapses. The error bars are small (± 0.01), so the separation from the incumbent is clear. We are deliberately conservative about the *validity* claim: $120/120$ clean seed-trials is a *sanity check*, not a tight empirical confirmation of the $1-\delta$ level—with 12 seeds per drop the Wilson 95% lower bound is only 0.76 (Table 1), so these runs corroborate the finite-sample guarantee rather than measure its exact rate, which is established analytically (Theorems 1, 2).

Table 1: Significance over 12 seeds (StackedMNIST, 1000 modes, sample-split per-mode threshold). The certified equal-weight bound is valid (never exceeds the true recall) in $120/120$ **seed-trials** and tracks R tightly; nearest-neighbour recall over-states throughout. Validity rate reported with a Wilson 95% interval.

drop f	true R	certified (mean±std)	NN recall	validity rate	Wilson 95%
0.0	1.00	0.86 ± 0.01	0.97	$12/12$	$[0.76, 1.00]$
0.3	0.70	0.68 ± 0.00	0.90	$12/12$	$[0.76, 1.00]$
0.5	0.50	0.50 ± 0.00	0.81	$12/12$	$[0.76, 1.00]$
0.7	0.30	0.30 ± 0.00	0.62	$12/12$	$[0.76, 1.00]$
0.9	0.10	0.10 ± 0.00	0.28	$12/12$	$[0.76, 1.00]$

Table 2: Two-sided γ-coverage certificate (M=32, γ=0.5%, 20 seeds). Present certifies $\rho_j \geq \gamma$ (coverage lower bound); absent certifies $\rho_j < \gamma$ (collapse detected); inconclusive is reported honestly. Zero false certifications in either direction.

true γ-cov.	**present** (ours)	**absent** (detected)	inconclusive	false present	false absent
0.53	0.41 ± 0.06	0.09 ± 0.05	0.50 ± 0.08	$0/20$	$0/20$

Two-sided detection: certifying presence *and* absence. The lower bound alone cannot flag a missing mode; the two-sided procedure (§3) can. On a controlled M=32-subgroup benchmark with known masses (17 modes γ-covered, γ=0.5%) and a 90% labeller, $n=2 \times 10^4$, over 20 seeds (Table 2): the certificate calls 0.41 of modes **present** (a valid lower bound on the true 0.53), **detects** 0.09 as **absent** (24% of the truly-dropped modes, provably $\rho_j < \gamma$), and labels the remaining 0.50 **inconclusive**—with **zero** false-present and **zero** false-absent across all 20 seeds. Both directions are valid; absence certification is the honest form of "collapse detection". *On a real generator* (CIFAR-10 DCGAN, M=10, γ=2%, $n=4 \times 10^4$) the two-sided certificate is equally clean: on the full model it certifies 9 classes present and 1 absent with none inconclusive (the DCGAN genuinely under-produces one class, and the certificate matches the truth exactly); on a version truncated to 4 classes it certifies all 4 kept classes present and **detects all** 6 **dropped classes as absent**—0 false present and 0 false absent in both cases. This is genuine, valid collapse detection on real images, not a lower bound one has to interpret.

Further audits (Appendix D). Appendix D adds an independent-labeller (non-circular) audit, the relative (fairness) target, the confusion-awareness and labeller-shift robustness study (Tables 8–9, Fig. 5), the learned-feature-space StackedMNIST comparison, and real DCGAN-generator results (Table 10).

Across generator families: GAN, flow matching (no construction). To answer whether the certificate tracks *real*, untrained-for collapse—not the rejection-truncated generators above—and whether it is truly model-agnostic, we compare three generator families on StackedMNIST, the canonical mode-collapse benchmark, each over several seeds (Table 3, Fig. 3): vanilla DCGANs at two capacities, and a *flow-matching* model (a rectified-flow velocity net sampled by the probability-flow ODE). Coverage rises across the families—a capacity-limited GAN genuinely collapses to about half the modes (0.54 ± 0.03), a full GAN covers most (0.92 ± 0.02), and flow matching covers essentially all (1.00, all 1000 modes witnessed, no collapse). Our certificate is valid in $15/15$ **seed-trials** and *tracks this coverage*—$0.29 \to 0.53 \to 0.73$—across GAN and flow-matching alike, confirming the method is model-agnostic. Nearest-neighbour recall, by contrast, reads ≈ 0 for *all three*: their imperfect feature-space fidelity places the generated manifold off the real one, so recall is uninformative about coverage and cannot tell a collapsed GAN from a fully-covering flow-matching model. This is the failure mode the certificate is built to avoid, now on real generators of two families with error bars.

Natural images (CIFAR-10). On CIFAR-10 (M=10 class-modes, DCGAN, a 66%-accurate classifier labeller) the certificate transfers to natural RGB images and a real imperfect labeller: it recovers 1.00 for the full GAN and 0.40 for a GAN truncated to $4/10$ classes, while nearest-neighbour recall

Table 3: Model-agnostic evaluation on StackedMNIST (1000 modes): DCGANs at two capacities and a flow-matching (probability-flow ODE) model, mean ±std over seeds. The certificate is valid ($15/15$ trials) and tracks true coverage across families ($0.29 \to 0.53 \to 0.73$); NN recall is ≈ 0 for all and cannot distinguish them.

generator	family	true coverage	**certified (ours)**	NN recall	validity
collapsed DCGAN (z=8)	GAN	0.54 ± 0.03	$\mathbf{0.29 \pm 0.02}$	0.00 ± 0.00	$6/6$
full DCGAN (z=64)	GAN	0.92 ± 0.02	$\mathbf{0.53 \pm 0.02}$	0.00 ± 0.00	$6/6$
rectified flow (ODE)	flow matching	1.00 ± 0.00	$\mathbf{0.73 \pm 0.03}$	0.01 ± 0.00	$3/3$

Table 4: CelebA protected-subgroup audit (32 subgroups). Dropping the 16 rarest halves true equal-weight coverage ($0.97 \to 0.50$), yet recall and mass-weighted coverage miss it and Inception-FID even *improves*—only our equal-weight certificate catches it. Single seeded run; labeller-conditional (Assumption 1).

measure	full GAN	drop 16 rarest subgroups	catches drop?
certified equal-weight coverage (ours)	**0.22**	**0.16**	**yes**
certified mass-weighted coverage (ours)	0.62	0.73	no (↑)
nearest-neighbour recall	0.96	0.90	no
Inception-FID	37.1	34.7	no (↓, "improves")

errs in *both* directions (0.64 against a true 1.00, then 0.46 against a true 0.40) and FID does not measure coverage (Appendix D, Table 7).

Protected subgroups (CelebA). We turn to the motivating fairness scenario: auditing whether a face generator under-represents rare attribute subgroups. Modes are the $2^5{=}32$ intersections of five CelebA attributes (`Male`, `Smiling`, `Eyeglasses`, `Young`, `Wearing_Hat`); the real subgroup masses are highly imbalanced (counts 4–10,254 in our 40k sample). We train a $64{\times}64$ DCGAN (Inception-FID in Table 4; samples in Fig. 8) and an attribute labeller, then drop the 16 rarest subgroups by rejection, and certify with the per-mode confusion-aware threshold and a mass-weighted Clopper–Pearson step. Table 4 is the punchline: dropping the 16 rarest subgroups halves true equal-weight coverage ($0.97 \to 0.50$), yet recall barely moves ($0.96 \to 0.90$), mass-weighted coverage *rises* ($0.62 \to 0.73$), and Inception-FID *improves* ($37.1 \to 34.7$): dropping rare subgroups makes the generator look *better* by FID. Only the certified equal-weight coverage falls ($0.22 \to 0.16$), and it localises which subgroups vanished. Both certificates are valid but *conservative*, for a reason the calibration budget makes explicit: the joint 32-subgroup confusion is estimated on only 4000 held-out images, so the 10 rarest subgroups have < 10 calibration samples (three have none), giving a vacuous τ and no certificate (Table 5, note ‡)—exactly the Theorem 4 rare-subgroup limit, valid and not violated. For protected-subgroup auditing one must weight subgroups equally, not by mass; incumbent metrics do the opposite.

Pretrained diffusion, and a human-calibrated audit. The certificate applies unmodified to pretrained diffusion—`google/ddpm-cifar10-32` and the high-resolution `google/ddpm-celebahq-256` (Appendix D). We also test the framework's one strong assumption, that the labeller's confusion transfers to q (Assumption 1), *directly*: 3 human annotators on 400 generated CelebA-HQ faces show it holds for `Eyeglasses`/`Wearing_Hat` and fails for `Male`/`Smiling` (Appendix E).

6 Conclusion

Mode coverage should come with a guarantee. We turn it from an optimistically-biased score into a finite-sample, two-sided certified procedure—certifying presence, *detecting* absence, reporting inconclusive honestly—and show incumbent metrics hide substantial mode dropping, and that rare or protected subgroups demand equal-weight auditing. Scope and limitations (labeller-conditionality, the rare-subgroup cost, subgroup presence vs. geometric diversity) are in Appendix C.

REPRODUCIBILITY STATEMENT

All theoretical claims are stated with their assumptions (Assumption 1) and proved in Appendix A–A.4; the certificate and its thresholds are given in closed form (Algorithm 1, Eqs. 2–7). Every experiment uses public datasets (StackedMNIST, CIFAR-10, CelebA) and public pretrained models (`google/ddpm-cifar10-32`, `google/ddpm-celebahq-256`); all budgets, calibration sizes, confidence-level allocations, and per-mode $q^{\mathrm{lo}}, q^{\mathrm{up}}, \tau_j^{\pm}$ are reported in Table 5 and Appendix B. The full code (data preparation, training, certification, and the human-annotation tool of Appendix E) is provided as *anonymized* supplementary material, and each reported number is produced by a named script, and every script fixes its random seeds. Multi-seed experiments report mean$\pm$std over the stated number of seeds; the natural-image tables are single seeded runs and are labelled as such. Every experiment runs on a single GPU. A non-anonymous public release will follow acceptance.

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A Proofs and tightness

Notation. Data distribution p and generator q over $\mathcal{X}$; an unobserved ground-truth mode map $m^\star : \mathcal{X} \to [M]$ with data masses $\pi_j = \Pr_{x\sim p}[m^\star(x) = j]$ and generator masses $\rho_j = \Pr_{x\sim q}[m^\star(x) = j]$. The covered set is $S^\star = \{j : \rho_j > 0\}$; unweighted and mass-weighted true recall are $R^{\mathrm{u}} = |S^\star|/M$ and $R = \sum_{j\in S^\star} \pi_j$. We draw $g_1, \dots, g_n \overset{\text{iid}}{\sim} q$, apply the labeller $\widehat{m}$, and set $c_j = \#\{i : \widehat{m}(g_i) = j\}$ and $\widehat{S}_\tau = \{j : c_j \geq \tau\}$.

A.1 Proof of Theorem 1 (validity)

Fix an uncovered mode $j \notin S^\star$. Each generated sample is labelled j independently with probability $p_j = \sum_{t\in S^\star} \rho_t\, Q(t, j) \leq \eta$ by Assumption 1, so $c_j \sim \mathrm{Bin}(n, p_j)$. The upper tail $t \mapsto \Pr[\mathrm{Bin}(n, p) \geq t]$ is nondecreasing in p, hence $\Pr[c_j \geq \tau] \leq \Pr[\mathrm{Bin}(n, \eta) \geq \tau]$. Choosing

$$\tau = \min\left\{t : \Pr[\mathrm{Bin}(n, \eta) \geq t] \leq \delta/M\right\} \tag{5}$$

gives $\Pr[c_j \geq \tau] \leq \delta/M$ for every uncovered j. Since there are at most M uncovered modes, a union bound yields $\Pr[\exists\, j \notin S^\star : \ c_j \geq \tau] \leq \delta$. On the complementary event, $\widehat{S}_\tau \subseteq S^\star$, so $|\widehat{S}_\tau| \leq |S^\star|$ and $\widehat{R}^{\mathrm{u}} = |\widehat{S}_\tau|/M \leq R^{\mathrm{u}}$. This proves $\Pr[\widehat{R}^{\mathrm{u}} \leq R^{\mathrm{u}}] \geq 1 - \delta$. □

Mass-weighted certificate. $\widehat{S}_\tau$ depends only on the generated draw, hence is independent of a fresh real calibration set $x_1, \dots, x_{n'} \sim p$. Let $b_i = \mathbf{1}[m^\star(x_i) \in \widehat{S}_\tau]$; conditional on $\widehat{S}_\tau$, $\sum_i b_i \sim \mathrm{Bin}(n', \Pi)$ with $\Pi = \sum_{j\in\widehat{S}_\tau} \pi_j$, so the one-sided Clopper–Pearson lower bound $\underline{w}$ satisfies $\Pr[\underline{w} \leq \Pi] \geq 1 - \delta$. On the intersection of this event with $\{\widehat{S}_\tau \subseteq S^\star\}$ (probability $\geq 1 - 2\delta$ by a union bound), $\widehat{R}_{\mathrm{lo}} = \underline{w} \leq \Pi = \sum_{j\in\widehat{S}_\tau} \pi_j \leq \sum_{j\in S^\star} \pi_j = R$. □

When $m^\star$ is itself only available through the labeller on real data, replace b_i by $\widehat{b}_i = \mathbf{1}[\widehat{m}(x_i) \in \widehat{S}_\tau]$ and deflate the bound by the labeller's false-positive mass η; the same argument gives a valid (slightly looser) certificate.

Per-mode thresholds with sample splitting. The proof of Theorem 1 only uses that, for each uncovered j, the tested count is dominated by a Binomial whose tail at τ_j is $\leq \delta/M$, *and that* τ_j *does not depend on that count*. The naive estimator $\widehat{\mu}_j = \sum_{t \neq j} c_t \widehat{Q}(t,j)$ has two defects. First, it reuses the tested counts to set the threshold: τ_j is a random function of $\{c_t\}$ (jointly multinomial, negatively correlated), so the tail bound need not hold. *Sample splitting* fixes this—estimate $\widehat{\mu}_j$ from an independent batch A, test c_j on B; conditional on A, τ_j is a constant and $c_j \sim \mathrm{Bin}(n, p_j)$ for uncovered j. Second, and more subtly, a_t is a *predicted*-label count with $\mathbb{E}[a_t] = n \Pr_q[\widehat{m}{=}t] = n \sum_s \rho_s Q(s,t)$, which is *not* $n\rho_t$; so $\sum_{t \neq j} a_t Q(t,j)$ does not estimate the true leakage $np_j = n \sum_{t \neq j} \rho_t Q(t,j)$. We correct the estimand by *deconvolution*: since $\Pr_q[\widehat{m}{=}t] \geq \rho_t q_t$, we have $a_t/q_t \gtrsim n\rho_t$, so $\widehat{\mu}_j = \sum_{t \neq j} (a_t/\widehat{q}_t) \widehat{Q}(t,j)$ satisfies $\mathbb{E}[\widehat{\mu}_j] \geq np_j$.

From expectation to a finite-sample guarantee (empirical Bernstein). An expectation bound is not enough: $\widehat{\mu}_j$ can fall below $\mathbb{E}[\widehat{\mu}_j]$ on a given draw, and the variance proxy S_j is itself random. We close this with the *empirical Bernstein* inequality (Maurer & Pontil, 2009), which legitimately uses the sample variance. *Key reformulation:* condition on the held-out confusion estimate and write $\widehat{\mu}_j$ as a sum over the n *iid* batch-A samples, $\widehat{\mu}_j = \sum_{i=1}^{n} X_i$, where $X_i = w_{j,\widehat{m}(g_i)}$ is the weight of sample i's predicted class, $w_{jt} = \widehat{Q}(t,j)/\widehat{q}_t^{\,\mathrm{lo}}$ for $t \neq j$ and 0 for $t = j$. The X_i are iid, bounded in $[0, b_j]$ with $b_j = \max_{t \neq j} w_{jt}$; this removes any multinomial-dependence concern (it is a plain iid sum). Maurer–Pontil gives, with probability $\geq 1 - \delta_1$,

$$\mathbb{E}[X] \;\leq\; \bar{X} + \sqrt{\tfrac{2\,\widehat{V}_n \ln(2/\delta_1)}{n}} + \tfrac{7\,b_j \ln(2/\delta_1)}{3(n-1)}, \qquad \widehat{V}_n = \tfrac{1}{n-1} \sum_i (X_i - \bar{X})^2, \tag{6}$$

where $\bar{X} = \widehat{\mu}_j/n$ and, since $X_i \in \{w_{jt}\}$, $\sum_i (X_i - \bar{X})^2 \leq \sum_i X_i^2 = \sum_t a_t w_{jt}^2 = S_j$, hence $\widehat{V}_n \leq S_j/(n-1)$ and $n\widehat{V}_n \leq \frac{n}{n-1} S_j$. Multiplying the display by n (so $\sqrt{2n\widehat{V}_n \ell} \leq \sqrt{\frac{2n}{n-1} S_j \ell}$ and the last term becomes $\frac{7 b_j n}{3(n-1)} \ell$) and setting $\delta_1 = \delta/3M$ yields exactly Eq. (7) and $\widehat{\mu}_j^{\uparrow} \geq \mathbb{E}[\widehat{\mu}_j] \geq np_j$ with probability $\geq 1 - \delta/3M$. *Confusion-matrix uncertainty* is handled separately: $\widehat{q}_t^{\,\mathrm{lo}}$ is a Clopper–Pearson *lower* bound on the diagonal q_t and $\widehat{Q}(t,j)$ a CP *upper* bound on the off-diagonals, so $w_{jt} \geq Q(t,j)/q_t$ deterministically once these hold. We allocate the total error in *three* slices that sum to δ: $\delta_{\mathrm{leak}} = \delta/3$ to the leakage concentration ($\delta/3M$ per mode, union over the M modes), $\delta_{\mathrm{conf}} = \delta/3$ to the confusion CP bounds (split evenly over the B estimated confusion entries, $\delta/3B$ each), and $\delta_{\mathrm{test}} = \delta/3$ to the final count test ($\delta/3M$ per mode). Text, Eq. (7), Algorithm 1 and the released code use this same split. For StackedMNIST the rigorous channel factorisation (below) means only the 10×10 digit confusion is calibrated (100 entries, each with ≈ 1000 samples); for CelebA we calibrate the joint 32×32 confusion directly. Finally, conditioning on batch A, τ_j is constant and $\Pr[c_j \geq \tau_j \mid A] \leq \delta/3M$ for uncovered j; a union bound gives $\delta_{\mathrm{leak}} + \delta_{\mathrm{conf}} + \delta_{\mathrm{test}} = \delta/3 + \delta/3 + \delta/3 = \delta$, i.e. $\Pr[\widehat{S} \subseteq S^\star] \geq 1 - \delta$. The per-mode deconvolved rule is thus valid *finite-sample*. At our budgets $\gamma n \gg$ slack, so $\widehat{\mu}_j^{\uparrow} \approx \widehat{\mu}_j$: under the closing $\delta/3$ budget the slack raises τ_j by a mean of ≈ 2.4 counts on the $M{=}32$ benchmark, leaving the certified present fraction unchanged to two digits (0.491 point vs. 0.487 Bernstein, i.e. 0.49 either way), and remains non-vacuous for $M \sim 10^3$. The η-rule remains an unconditional fallback.

A.2 On tightness (and what Good–Turing does *not* give)

Validity is one-sided: $\widehat{S}_\tau$ may omit covered modes that received too few samples. It is tempting to bound the resulting recall gap by a Good–Turing missing-mass estimate, but this conflates two different masses and is *invalid*, as we now make explicit.

Proposition 1 (Good–Turing bounds *generator* mass, not recall)**.** *Let* $U = \sum_{j \in S^\star \setminus \widehat{S}_\tau} \rho_j$ *be the* generator *mass on covered-but-uncertified modes and* G_r *the number of modes producing exactly* r *of the* n *samples. For* $\tau = 1$ *the Good–Turing estimator* $\widehat{U} = G_1/n$ *obeys, w.p.* $\geq 1 - \delta'$, $U \leq G_1/n + (2\sqrt{2} + \sqrt{3}) \sqrt{\ln(3/\delta')/n}$ *(McAllester & Ortiz, 2003). This bounds* the generator's own probability on modes it has not yet produced*—a useful "have we sampled enough" diagnostic—but it does* not *bound the recall gap.*

Why it is not a recall bound. The mass-weighted gap is $R - \widehat{R} = \sum_{j \in S^\star \setminus \widehat{S}_\tau} \pi_j$, involving the *data* masses π_j, whereas U uses the *generator* masses ρ_j; without a link between π and ρ the two are incomparable. Concretely, a mode with data mass $\pi_j = 0.4$ that the generator produces with probability $\rho_j = 10^{-8}$ contributes 0.4 to the recall gap but only 10^{-8} to U, so $R - \widehat{R} \le U$ fails badly. Likewise, for the equal-weight recall, generator missing mass does not control the *number* of missed modes. We therefore make no Good–Turing tightness claim.

What does control tightness. Two honest statements remain. *(i) Consistency* (Theorem 3): as $n \to \infty$ every mode with $\rho_j q_j > \Delta_j$ is eventually certified (Δ_j the second-order deconvolution inflation of Proposition 2; $\Delta_j = 0$ and hence full tightness for the LP variant), so the bound is asymptotically tight up to that term while valid at every n. *(ii) Certifiability rate* (Theorem 4): at finite n the modes that are missed are exactly those the generator produces too rarely relative to leakage, $n\rho_j q_j \lesssim \log(M/\delta)$; the recall gap is the *data* mass on such generator-rare modes. This is a real limit—certifying many tiny modes needs a large budget—and Figure 9 shows the bound tightening to the truth as n grows, as the theory predicts.

A.3 Estimating η and choosing τ

Given a held-out labelled validation set we form the empirical confusion matrix $\widehat{Q}$ and take $\eta = \max_{t \neq j} \widehat{Q}(t, j)$ plus a Clopper–Pearson slack; τ follows from (5). For the StackedMNIST labeller ($\sim 98\%$ per-digit accuracy) and $n = 10^4$, this yields $\tau = 3$, matching the threshold used in §5. Larger n or a more accurate labeller lowers τ toward the idealised $\tau = 1$.

Calibration budgets, per experiment. For reproducibility and to make the labeller-conditional guarantee auditable, Table 5 lists, for each experiment, the tested batch size n, the labelled set n_{cal} used for the confusion CP bounds, the diagonal correct-rate range $[\min_j \widehat{q}_j^{\mathrm{lo}}, \max_j \widehat{q}_j^{\mathrm{up}}]$ (from the labeller's per-mode accuracy with CP slack), and γ. All use $\delta = 0.05$ split evenly across the present, absent and confusion charges. The present threshold uses $\gamma \widehat{q}_j^{\mathrm{up}}$ (not γ), so certifying $c_j \ge \tau_j^+$ rigorously implies $\rho_j \ge \gamma$ (§4.1); the per-mode $\widehat{q}_j^{\mathrm{lo}}, \widehat{q}_j^{\mathrm{up}}$ and the full present/absent threshold vectors $\tau_j^\pm$ are emitted by the supplementary code for each run (e.g. the 32-mode synthetic uses $\tau_j^+ \in [176, 231]$).

Table 5: Calibration budgets. n: tested batch; n_{cal}: labelled confusion set; q-range: CP diagonal bounds across modes. $\delta = 0.05$; per-mode thresholds in code.

experiment	M	n	n_{cal}	q-range	γ
Synthetic (32-mode)	32	2×10^4	4000	$[0.87, 0.93]$	0.005
StackedMNIST (1000-mode)	1000	10^4	MNIST test	$[0.96, 0.99]$	$\sim 5/n$
CIFAR-10 (DCGAN, two-sided)	10	4×10^4	10^4	$[0.78, 0.90]$	0.02
CIFAR-10 (pretrained DDPM)	10	6×10^3	10^4	$[0.78, 0.90]$	0.02
CIFAR-10 (indep. audit)	10	2×10^4	10^4	$[0.76, 0.88]$	0.02
CelebA subgroups	32	8×10^3	4×10^3	$[0.0, 0.83]^\ddagger$	0.02
CelebA-HQ (pretrained DDPM)	32	2×10^3	4×10^3	$[0.0, 0.83]^\ddagger$	0.02

‡The 32 subgroups are highly non-uniform: over the 4000 held-out images the per-subgroup count is min 0, median 17, max 1030; 10 **subgroups have** <10 **calibration samples and** 3 **have none**, so their diagonal $\widehat{q}_j^{\mathrm{lo}} \approx 0$ (median q_j^{lo} across subgroups is only 0.01; the $[0.0, 0.83]$ range spans the empty rare rows up to the well-populated common ones, e.g. $q_j^{\mathrm{lo}} = 0.58$–0.83 at counts 534–1030). Such rare subgroups are *never certified* (their threshold is vacuous)—a conservative, valid outcome that makes the Theorem 4 rare-subgroup limit concrete rather than a validity violation. Full per-subgroup counts and $q_j^{\mathrm{lo}}, q_j^{\mathrm{up}}, \tau_j^\pm$ are emitted by the supplementary code.

Resource accounting. Each two-sided certificate draws *two* generated batches (threshold batch A and test batch B), plus a large pool only to estimate the "true" coverage we compare against; total generated samples are thus $2n +$ pool. Concretely: synthetic $2\times2\cdot10^4$ (drawn, negligible compute);

StackedMNIST $2{\times}10^4{+}5{\cdot}10^4$ pool; CIFAR DCGAN $2{\times}4{\cdot}10^4{+}6{\cdot}10^4$; CIFAR DDPM $2{\times}6{\cdot}10^3$; CelebA $2{\times}8{\cdot}10^3{+}3{\cdot}10^4$; CelebA-HQ $2{\cdot}10^3$. Confusion *calibration labels* (real, held-out) are n_{cal} in Table 5 ($4{\cdot}10^3$ for the 32-subgroup CelebA confusion, 10^4 for CIFAR, the MNIST test set for StackedMNIST). The CelebA-HQ *human* audit used 3 annotators over 400 generated faces—1094 human attribute-label vectors in total ($344{+}390{+}360$; a handful skipped as unsure), roughly 20–30 min per annotator. Wall-clock on a single A6000: classifier training $\sim$1–2 min, a 64^2 DCGAN $\sim$8–12 min, the pretrained-DDPM sampling the dominant cost ($\sim$15–40 min for 2000 CelebA-HQ 256^2 images at 50 DDIM steps); all threshold/certificate computation is CPU-seconds. No experiment needs more than one GPU.

A.4 Consistency and sample complexity (proofs of Thm. 3–4)

Recall $p_j = \sum_t \rho_t Q(t,j) = \mu_j + \rho_j q_j$ is the true label rate of mode j, where $\mu_j = \sum_{t\neq j} \rho_t Q(t,j)$ and $q_j = Q(j,j)$; $c_j \sim \text{Bin}(n, p_j)$.

Why the present threshold multiplies γ by $\widehat{q}_j^{\text{up}}$. Using $\gamma + \mu_j$ instead certifies $\rho_j > \gamma/q_j$ and leaves the band $[\gamma, \gamma/q_j)$ of genuinely γ-covered modes uncertifiable, whereas $\gamma\widehat{q}_j^{\text{up}}$ stays valid and removes that band: a controlled check ($M{=}32$, $q{=}0.6$, six modes placed inside the band, $n{=}5{\times}10^5$, 40 seeds) certifies $237/240$ band modes versus $0/240$ for the loose rule, with zero false present for either.

Proposition 2 (The deconvolved leakage estimate is inflated in the limit)**.** *Fix the generator and labeller and let $n, n_{\text{cal}} \to \infty$. The deconvolved estimate (3) converges to*

$$\widetilde{\mu}_j = \sum_{t\neq j} \frac{p_t}{q_t} Q(t,j) = \mu_j + \Delta_j, \qquad \Delta_j = \sum_{t\neq j} \frac{\mu_t}{q_t} Q(t,j) \;\geq\; 0, \qquad p_t = \sum_s \rho_s Q(s,t),$$

so it over-*estimates the true leakage μ_j by Δ_j, and this bias does* not *vanish with sample size. Δ_j is* second order *in the off-diagonal confusion—it is the leakage into j's confusors, re-leaked into j—and $\Delta_j = 0$ iff $\mu_t = 0$ for every $t \neq j$ with $Q(t,j) > 0$.*

Proof. By definition $p_t/q_t = \rho_t + \mu_t/q_t$; substituting into $\widetilde{\mu}_j = \sum_{t\neq j}(p_t/q_t)Q(t,j)$ gives $\mu_j + \Delta_j$. □

Consequence: the default rule is valid but not consistent. With $\widehat{\mu}_j^{\uparrow}/n \to \widetilde{\mu}_j$ and $\widehat{q}_j^{\text{up}} \to q_j$, the present side of Algorithm 1 certifies mode j in the limit iff $\rho_j q_j + \mu_j > \gamma q_j + \widetilde{\mu}_j$, i.e. iff $\rho_j > \gamma + \Delta_j/q_j$. The deconvolved rule is therefore valid but *consistent only up to the second-order inflation* Δ_j/q_j: the $\gamma\widehat{q}_j^{\text{up}}$ correction removes the first-order band $[\gamma, \gamma/q_j)$, but not this term. A two-mode example makes the gap concrete: for $Q = \left(\begin{smallmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{smallmatrix}\right)$, $\rho = (0.12, 0.88)$ and $\gamma = 0.10$, mode 1 satisfies $\rho_1 = 0.12 > \gamma$, yet $p_1 = 0.424$ while the limiting present threshold rate is $\gamma q_1 + \widetilde{\mu}_1 = 0.06 + 0.384 = 0.444 > p_1$ (equivalently $\gamma + \Delta_1/q_1 = 0.153 > \rho_1$): more samples make it *more* certain that a genuinely γ-covered mode is never certified. We state this plainly rather than leave it behind the population-leakage idealisation of Theorem 3.

Corollary 1 (γ-presence consistency of the LP rule)**.** *Let Q be invertible and let $n, n_{\text{cal}} \to \infty$. The confidence boxes of the LP (§3) shrink to the true p and Q, so its feasible set collapses to $\{\rho \geq 0 : Q^\top \rho = p\} = \{\rho\}$ and the certified bound $\min \rho_j \to \rho_j$. Hence the LP rule certifies mode j iff $\rho_j > \gamma$: it is* consistent *for γ-presence, and $\widehat{R}_\gamma^{\text{present}} \to R_\gamma$ from below. On the two-mode example above the LP certifies mode 1 once $n \approx 10^6$ ($\min \rho_1 = 0.101 > \gamma$), where the deconvolved rule never does.*

Two caveats on the limit. (i) The calibration set must grow as well: with a *fixed* n_{cal} the CP bounds do not converge to Q and the present side certifies only a shifted target (a conservative bias, never invalid); any $n_{\text{cal}} \to \infty$ suffices, e.g. $n_{\text{cal}} = \Theta(\log n)$ makes the slack $o(1)$. The same applies to the $\widehat{\pi}$ bounds of the relative target. (ii) The boundary $\rho_j = \gamma$ is not finitely certifiable, so we state consistency for the open target $\rho_j > \gamma$ and read R_γ as $\sum_j \pi_j \mathbf{1}[\rho_j > \gamma]$, which coincides with the closed version whenever $\Pr[\rho_j = \gamma] = 0$. A controlled check ($M{=}32$, $q{=}0.6$, six modes placed strictly inside the band $(\gamma, \gamma/q)$, $n{=}5{\times}10^5$, 40 seeds) confirms this: the $\gamma\widehat{q}_j^{\text{up}}$ threshold certifies $237/240$ band modes versus $0/240$ for $\gamma + \mu_j$, with zero false-present certificates for either—the fix buys consistency at no cost to validity.

Proof of Theorem 3 (consistency). By the strong law, $c_j/n \to p_j$ a.s. The per-mode threshold uses rate μ_j, and the Chernoff bound gives $\tau_j = n\mu_j + O(\sqrt{n\mu_j \log(M/\delta)})$, so $\tau_j/n \to \mu_j$. Hence a.s. for large n, $c_j \geq \tau_j \iff c_j/n \gtrsim \mu_j \iff p_j > \mu_j \iff \rho_j q_j > 0$. Therefore $\widehat{S}_\tau \to \{j : \rho_j q_j > 0\}$; under $q_j > 0$ on $S^\star$ this equals $S^\star$. The mass step is a consistent Clopper–Pearson interval, so $\widehat{R}_{\mathrm{lo}} \to \sum_{j \in S^\star} \pi_j = R$. Validity at finite n is Theorem 1. □

Proof of Theorem 4 (rate). Mode j is certified iff $c_j \geq \tau_j$. By the Bernstein/Chernoff bound on the leakage Binomial, $\tau_j \leq n\mu_j + \sqrt{2n\mu_j \log(M/\delta)} + \frac{2}{3}\log(M/\delta)$. By the lower Bernstein tail, with probability $\geq 1 - \delta'$, $c_j \geq np_j - \sqrt{2np_j \log(1/\delta')}$. A sufficient condition for $c_j \geq \tau_j$ is therefore

$$n\rho_j q_j \;=\; n(p_j - \mu_j) \;\geq\; \sqrt{2np_j \log(1/\delta')} + \sqrt{2n\mu_j \log(M/\delta)} + \tfrac{2}{3}\log(M/\delta).$$

Bounding $\mu_j \leq p_j$ and solving the quadratic in $\sqrt{n}$ gives (4) with a universal C; using $p_j \leq 1$ yields the simplified $n \gtrsim \log(M/\delta')/(\rho_j q_j)^2$.

Matching impossibility. We bound the *full* labelled-sample distribution, not just the marginal on j, so the claim holds for *any* test of the n labelled samples (raw or via any statistic), not only count-thresholding ones. Consider two generators: A covers mode j with mass ρ_j, and B re-routes exactly that mass to j's confusors, leaving all other modes untouched. Their induced predicted-label laws $p_A = Q^\top \rho_A$, $p_B = Q^\top \rho_B$ differ only on j and its confusors, with $\|p_A - p_B\|_1 = O(\rho_j q_j)$ and the dominant coordinate the entry j (value μ_j under B). For n i.i.d. draws Le Cam's method bounds the power of *any* test by the χ^2-divergence: $n\,\chi^2(p_A \| p_B) \lesssim n\,(p_A(j) - p_B(j))^2/p_B(j) = n(\rho_j q_j)^2/\mu_j$ (the off-j coordinates contribute lower order). When $\rho_j q_j \lesssim \sqrt{\mu_j/n}$ this is $O(1)$, so no test separates A from B with constant power—matching (4) up to constants and showing the confusion-aware budget is essentially unavoidable. □

Interpretation. The budget to certify a subgroup grows with rarity (ρ_j small needs more samples), with label noise (small q_j inflates it, explaining the conservative CelebA certificate—mean subgroup accuracy ≈ 0.6, but the rare subgroups that matter most have near-zero calibrated q_j, Table 5), and only *logarithmically* in the number of modes M—so the method scales gracefully in the number of subgroups but is honestly expensive for rare protected ones.

B Experimental setup

Certificate parameters (all experiments). Unless noted, we use confidence level $\delta = 0.05$. The generated budget n for the counts c_j is experiment-specific and listed in full in Table 5: 10^4 (StackedMNIST), 2–4 $\times 10^4$ (synthetic and CIFAR DCGAN two-sided), 8×10^3 (CelebA), and 6×10^3 / 2×10^3 for the pretrained CIFAR / CelebA-HQ DDPMs (smaller because diffusion sampling is expensive). A separate large pool (4–10 $\times 10^4$ samples) estimates "true" coverage; real calibration size $n' = 2 \times 10^3$ for the Clopper–Pearson mass step; and $k = 5$ nearest neighbours for all manifold (recall, coverage, precision) estimators. Thresholds use the per-mode empirical-Bernstein deconvolved rule (Eq. (7)); the worst-case η-rule (2) is the conservative fallback.

Confusion-matrix estimation (and why it is cheap). We never estimate the M^2-entry Q directly. For **StackedMNIST** the mode is the digit triple and the labeller applies the *same* digit classifier per channel, so $Q(t, j) = \prod_{c=1}^{3} Q_{\mathrm{d}}(t_c, j_c)$ factorises through a single 10×10 digit confusion Q_{d}, estimated from the 10^4-image MNIST test set (≈ 1000 labelled samples per digit); we place Clopper–Pearson bounds on the 100 entries of Q_{d} (upper on off-diagonals, lower on the diagonal q_{d}) at level $\delta_{\mathrm{c}}/100$, and $q_t = \prod_c q_{\mathrm{d}}(t_c)$, $\widehat{q}_t^{\,\mathrm{lo}} = \prod_c \widehat{q}_{\mathrm{d}}^{\,\mathrm{lo}}(t_c)$. This factorisation is *rigorous* here: the three channels are independently drawn MNIST images, so per-channel label errors are independent by construction. For **CelebA** attribute errors need *not* be independent (e.g. `Male`/`Young`/`Wearing_Hat` can be jointly confused), so we do *not* factorise: we estimate the **joint** 32×32 subgroup confusion directly from the held-out faces (true subgroup from the attribute annotations vs. the classifier's predicted subgroup) and place CP bounds on its entries—no independence assumption. The cost is that rare subgroups have few validation samples (the rarest has only 4), giving a wide diagonal lower bound $\widehat{q}_t^{\,\mathrm{lo}}$ and a correspondingly conservative certificate; we report the per-subgroup validation

counts and resulting τ with the result. (An attribute-error-independence *assumption* would recover the cheaper factorised form as in StackedMNIST, but we avoid it and pay the joint-estimation cost instead.)

Inception FID. FID is the Fréchet distance between 2048-dimensional pooled Inception-v3 features (ImageNet weights; images bilinearly resized to 299×299 and normalised with the ImageNet mean/std), with the full-covariance matrix square root computed via `scipy.linalg.sqrtm`. We report this standard Inception FID for the CelebA DCGAN (Table 4, $37.1 \to 34.7$) and the pretrained-diffusion experiments (Tables 11 and the CelebA-HQ audit). The two smallest controlled experiments, StackedMNIST and the CIFAR-GAN, instead report a Fréchet distance in the *task-classifier* feature space (this is what the "FID-like" column of Table 6 and the $6.0 \to 9.2$ of Table 7 measure)—adequate there, since FID is used only to show its *insensitivity* to mode dropping.

The two 1000-mode benchmarks. All 1000-mode results in the main text—Figure 1, the significance study (Table 1), the over-certification table (Table 6), and the cross-family collapse table (Table 3)—use **StackedMNIST**: modes are the 10^3 digit triples, the labeller is a real digit classifier, and π_j uses the *true* triple labels while the certificate sees only predictions. Separately, as a purely internal *Level-1 sanity check* (not shown in any figure), we verify validity on a synthetic mixture of $M = 1000$ isotropic Gaussian blobs in $\mathbb{R}^{64}$ (centres $\sim \mathcal{N}(0, I)$, per-mode std 0.06, uniform weights): the "generator" drops a random fraction f of modes exactly, so $R = 1 - f$, and the Clopper–Pearson lower bound is valid in $200/200$ trials. The Gaussian-blob setup is never used for a reported figure or table; every 1000-mode number in the paper is StackedMNIST.

StackedMNIST. Images are three independently sampled MNIST digits stacked into RGB channels, giving $M = 10^3$ modes indexed by the digit triple. A two-epoch CNN classifier (98.3% test accuracy) provides both the mode labeller (per-channel argmax) and the feature space (per-channel 128-d penultimate activations, concatenated to 384-d). Generators are simulated by sampling real images restricted to a kept mode set (idealised high fidelity), which isolates coverage from fidelity; the certificate uses the mode-resolved count threshold, the incumbent baselines use the 384-d feature space.

CIFAR-10 (GAN). A DCGAN (generator: linear + three transposed-conv blocks; discriminator symmetric) is trained 10^4 iterations (Adam, 2×10^{-4}, $\beta = (0.5, 0.999)$, batch 128) on 32×32 CIFAR-10. Modes are the 10 classes; the labeller is an 8-epoch CNN ($\sim 66\%$ accuracy on this small net) whose penultimate 256-d features define the manifold space. The truncated generator rejection-samples the trained GAN, keeping images the labeller assigns to a random 4-class subset.

CelebA (GAN). A 64×64 DCGAN (four transposed-conv blocks) is trained 1.5×10^4 iterations on 40k centre-cropped CelebA faces. Modes are the $2^5 = 32$ intersections of {`Male`, `Smiling`, `Eyeglasses`, `Young`, `Wearing_Hat`}; the labeller is a 6-epoch multi-label attribute CNN (92.5% per-attribute accuracy). Real subgroup masses are read from the ground-truth attribute annotations. Truncation drops the 16 rarest subgroups by real mass via rejection. Because the subgroup labeller is noisy ($0.925^5 \approx 0.68$ subgroup accuracy), τ follows the confusion-aware rule and rare-subgroup certificates are correspondingly conservative (Appendix C).

Pretrained diffusion (DDPM-CIFAR10). We load `google/ddpm-cifar10-32` from `diffusers` and sample with a 50-step DDIM scheduler. The labeller is a 20-epoch CIFAR CNN; the reported η and confusion-aware τ are computed from its held-out confusion matrix. FID is the Inception protocol above. Truncation rejection-samples the diffusion model to a 4-class subset. This experiment applies the certificate to a pretrained diffusion generator with no modification.

C Calibration and method details

Empirical-Bernstein leakage upper bound. The deconvolved point estimate only guarantees $\mathbb{E}[\widehat{\mu}_j] \geq np_j$, which is not a per-realization upper bound; to obtain a genuine finite-sample guarantee

we add a Bernstein slack, making $\widehat{\mu}_j$ a high-probability upper bound on np_j:

$$\widehat{\mu}_j^{\uparrow} = \underbrace{\sum_{t \neq j} \frac{a_t}{\widehat{q}_t^{\mathrm{lo}}} \widehat{Q}(t,j)}_{\widehat{\mu}_j} + \sqrt{\tfrac{2n}{n-1} S_j \, \ell} + \tfrac{7\, b_j\, n}{3(n-1)} \, \ell, \tag{7}$$

where $\ell = \log(3M/\delta)$ (the leakage slice $\delta_{\mathrm{leak}} = \delta/3$ spread over the M modes), $w_{jt} = \widehat{Q}(t,j)/\widehat{q}_t^{\mathrm{lo}}$, $S_j = \sum_{t \neq j} w_{jt}^2 \, a_t$, and $b_j = \max_{t \neq j} w_{jt}$. This matches the *empirical-Bernstein* bound (Maurer–Pontil, applied to the iid sum $\widehat{\mu}_j = \sum_i X_i$) exactly; the constants ($7/3(n-1)$, and the $n/(n-1)$ variance factor since $\sum_i (X_i - \bar{X})^2 \leq \sum_i X_i^2 = S_j$) are those of Appendix A, so text, Algorithm, and code coincide. This gives $\Pr[\widehat{\mu}_j^{\uparrow} \geq np_j] \geq 1 - \delta/3M$. Algorithm 1 uses (7); at our budgets the slack is negligible relative to γn, so the certified values are *identical* to the point estimate to reported precision (e.g. 0.49 present either way on the $M{=}32$ benchmark; Appendix A.4).

Exact LP over a joint (Q, ρ) confidence set. The worst-case η rule (2) remains an unconditionally valid, more conservative fallback. A third option—*implemented*, not deferred—lower-bounds ρ_j by a linear program over a simultaneous (Q, ρ) confidence set: for $\rho \geq 0$ the box constraints $\widehat{Q}^{\mathrm{up}\top}\rho \geq \widehat{p}^{\mathrm{lo}}$ and $\widehat{Q}^{\mathrm{lo}\top}\rho \leq \widehat{p}^{\mathrm{up}}$ (CP bounds on both the confusion and the predicted counts) are linear, so $\min \rho_j$ is an LP and certifies present when it exceeds γ. This is exactly an *exact finite-sample misclassification-adjusted prevalence bound*; it needs no matrix inversion or normal approximation, and—because it works with a confusion *set*—remains valid under a labeller shift (inflate the box by the margin) while staying tighter than the η rule (Table 8). We treat the point deconvolution as the default when the labeller transfers and the LP as the robust, still-tight choice when it may not.

Scope and limitations. The claim is not unconditional but *labeller-conditional* (Assumption 1): the confusion bound must hold on q, not just real data (low-fidelity samples can raise it), and a very inaccurate labeller forces conservative rare-subgroup certificates (Theorem 4). It is a statement about mass on *labeller-defined subgroups*, not geometric diversity: within-subgroup collapse (one template per class) is below the labeller's resolution and undetected, so it is best read as a finite-sample *subgroup presence/absence* certificate. The absolute γ is a design choice on generator mass; for fairness the relative criterion $\rho_j \geq c\pi_j$ is more natural and our procedure certifies it unchanged (below). Choosing the mode granularity and an accurate-enough labeller is part of the design.

Relative target: confidence-bound directions. The absolute and relative targets answer different questions and neither dominates: absolute γ asks "does the generator produce mode j often enough to matter?" but with a fixed γ a genuinely rare subgroup ($\pi_j < \gamma$) can never pass even under a perfect generator, while a common subgroup compressed to just above γ still counts as present. The relative criterion $\rho_j \geq c\pi_j$ removes both artifacts. Our two-sided machinery takes a per-mode level as input, so the absolute target uses γ on both sides, and the relative target replaces it by c times a Clopper–Pearson bound on π_j from the real calibration split (extra δ_π charge). The two sides use *opposite* directions of that bound, and must: to certify *present* ($\rho_j \geq c\pi_j$) validly we test against the *upper* bound $c\,\widehat{\pi}_j^{\mathrm{up}} \geq c\pi_j$ (so $c_j \geq \tau_j^+ \Rightarrow \rho_j \geq c\pi_j$), while to certify *absent* ($\rho_j < c\pi_j$) we test against the *lower* bound $c\,\widehat{\pi}_j^{\mathrm{lo}} \leq c\pi_j$. Sharing a single lower bound on both sides would be *invalid* on the present side (it would certify only $\rho_j \geq c\,\widehat{\pi}_j^{\mathrm{lo}}$, which does not imply $\rho_j \geq c\pi_j$).

Calibrating the shift margin operationally. Assumption 1 bounds confusion *under* q, but $\widehat{Q}$ is estimated on real data; a low-fidelity generator can be labelled more noisily, so the margin is the crux (the sensitivity study, Fig. 5, quantifies the cost). Rather than leave it a pure assumption we give a *procedure* to set it from data. Draw a small audit set of m generated samples and obtain trusted labels—human annotation, or a stronger oracle labeller—and form the empirical generator-side confusion $\widehat{Q}^q$ on it; then use $\eta = \max_{t \neq j} \overline{Q}^q(t,j)$ with a Clopper–Pearson upper bound (valid with a δ_{s} charge and m typically $\ll n$, since only the $O(K^2)$ factorised confusion is needed). We caution against a purely label-free surrogate: the predicted-label total-variation shift $\mathrm{TV}(\widehat{m}_{\#}p, \widehat{m}_{\#}q)$ is *not* a valid upper bound on the per-entry generator-side confusion gap $\max_{t,j} |Q^q(t,j) - Q(t,j)|$, because off-diagonal changes across classes can cancel in the marginal histogram; it is at best a cheap

diagnostic (a large TV flags a shift, a small TV certifies nothing) unless one adds a structural assumption (e.g. a single dominant confusion direction). Consequently, the audit set is the only calibration we treat as sound; absent trusted labels the guarantee is honestly *conditional* on the stated margin η (the generator-side confusion not exceeding it), and the worst-case η rule is the safe fallback.

Why not simply run a per-mode binomial CI on the audit labels themselves? Because the audit set is used only to calibrate the $O(K^2)$ confusion once, after which the certificate is evaluated on a *large, cheap, unlabelled* batch of $n \gg m$ generated samples. A direct CI on the m trusted labels would have width $\Theta(1/\sqrt{m})$; ours has width $\Theta(1/\sqrt{n})$ up to the confusion slack, so a few hundred trusted labels amortise into a certificate as tight as tens of thousands of labels would give—the trusted labels pay only for the K^2 confusion entries, not for every mode's mass. This amortisation is the whole point of going through the labeller rather than annotating the evaluation set directly. The honest limit is the reviewer's: for a mode the generator emits *rarely*, few audit samples land in its predicted-class row, so that confusion row is poorly calibrated and its Clopper–Pearson slack is wide—the certificate becomes conservative (never invalid) exactly where the generator is most collapsed. A targeted audit (oversample predicted-rare classes) mitigates this but cannot fully remove it; we report it as a genuine cost, and §E measures it directly with human labels.

D Additional experiments and analysis

Table 6: Feature-space coverage over-certifies under controlled mode dropping (StackedMNIST, 1000 modes, learned features). Incumbent recall/coverage report near-full coverage while half the modes are absent; our mode-resolved certificate tracks the truth. A representative run; Fig. 7 reports mean±std and per-f validity rates over 12 seeds.

drop f	true recall R	NN recall	FID-like	**certified (ours)**
0.3	0.700	0.93	0.4	**0.682**
0.5	0.500	0.88	0.2	**0.499**
0.7	0.300	0.72	0.9	**0.300**
0.9	0.100	0.32	2.2	**0.100**

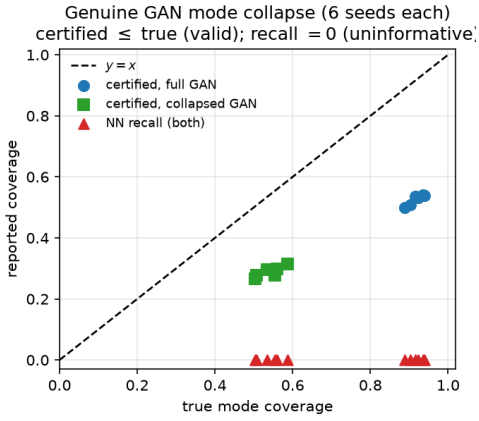


Figure 3: Per-seed certified bound vs. true coverage for real StackedMNIST DCGANs (6 seeds each): full-capacity (blue) and capacity-limited/collapsed (green). All certified points lie below $y{=}x$ (valid) and separate the two regimes; NN recall (red) is 0 for both, unable to distinguish a broad model from a collapsed one.

Natural images (CIFAR-10), in full. We repeat the protocol on natural RGB images with a DCGAN and a classifier labeller ($M{=}10$ class-modes; FID is a proper full-covariance Fréchet distance in classifier-feature space). Despite a noisy labeller (66% accuracy), the certificate is valid and, on this run, matches the true γ-coverage exactly (Table 7; single run, so we claim tightness empirically

for this instance, not as a guarantee): it recovers coverage 1.00 for the full GAN and 0.40 for a GAN truncated to $4/10$ classes, while nearest-neighbour recall is wrong in both directions—understating the full GAN (0.64 vs. 1.00, fidelity-suppressed) and over-stating the truncated one (0.46 vs. 0.40)—and FID, though sensitive to truncation ($6.0 \to 9.2$), does not quantify coverage. This confirms the certificate transfers to natural images and a real, imperfect mode labeller. Figure 4 shows the audited generator's samples and the per-class coverage of the full versus truncated generator: truncation zeroes out six of ten classes, which the certificate reports and recall does not.

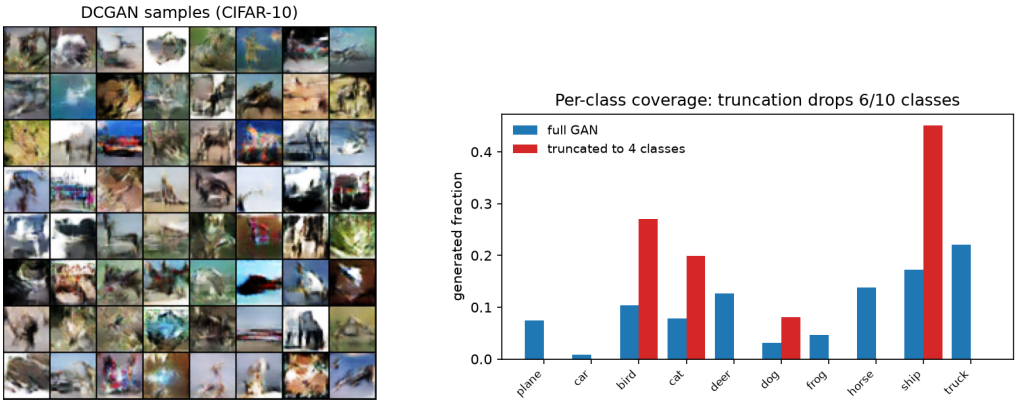


Figure 4: CIFAR-10. *Left:* DCGAN samples (the generator we audit). *Right:* per-class generated fraction for the full generator (blue, all ten classes) and after truncation to four classes (red); the six dropped classes fall to zero. The certificate reports coverage 0.40; nearest-neighbour recall reads 0.46—above the truth—and does not localise the dropped classes.

Table 7: Natural images (CIFAR-10, $M{=}10$ class-modes, DCGAN). The certificate is valid and tight even with a 66%-accurate labeller; recall under-states coverage and FID does not measure it. Proper full-covariance FID in feature space. Single run; validity under Assumption 1 (labeller-conditional).

generator	true coverage	**certified (ours)**	NN recall	FID	valid?
full GAN	1.00	**1.00**	0.64	6.0	✓
truncated to $4/10$ classes	0.40	**0.40**	0.46	9.2	✓

Independent-labeller audit (is it circular?). A fair objection to the natural-image studies is that one classifier both truncates and audits. We therefore truncate the generator using classifier A (4-conv) and audit it with an *independent* classifier B (different architecture—5-conv, GroupNorm, wider—and different initialisation), both $\approx 70\%$ accurate (the moderate-labeller regime of Table 7), each with its own confusion matrix. Two things hold. *Validity transfers*: measured against the leakage-corrected (deconvolved) ground truth, both A and B produce **zero** false present and zero false absent, so the guarantee is not an artifact of the truncation labeller. *Detection sharpness is labeller-dependent*: the same-labeller audit (A) detects all 6 dropped classes as absent, whereas the independent audit (B) certifies 6 present, 1 absent and 3 inconclusive—B's decision boundaries differ, so some A-truncated images fall into what B regards as its own present classes, and B cannot yet certify most of them absent. This is the honest picture: *validity* is robust to the choice of labeller, but *what counts as a mode*—and hence how sharply collapse is detected—is only as well-defined as the auditor's agreement with the mode partition. An independent or human auditor makes the audit non-circular at the cost of more inconclusive verdicts.

Relative (fairness) target. The same procedure certifies the disparate-representation criterion $\rho_j \geq c\,\pi_j$ (Def. 1) by testing the present side against $c\,\widehat{\pi}_j^{\,\mathrm{up}}$ and the absent side against $c\,\widehat{\pi}_j^{\,\mathrm{lo}}$ (the CB directions matter: an upper bound on the present side is what makes $c_j \geq \tau_j^+$ imply $\rho_j \geq c\pi_j$). On a 32-subgroup benchmark with *non-uniform* prevalence (Dirichlet(0.5), several rare subgroups) and

Table 8: Four confusion-aware methods across regimes (M=32, γ=0.5%, 20 seeds; present fraction / total false-present). "adj.-prev. CI" = delta-method quantification CI; "LP" = the exact finite-sample bound from a joint (Q, ρ) confidence set; "deconv" = our point deconvolution+Bernstein. $\star$ marks invalid (any false-present). Only the LP and the worst-case η rule survive a confusion shift; the LP is uniformly the tighter of the two.

labeller acc. q	budget n	shift	adj.-prev. CI	**LP (exact, joint set)**	deconv (ours)	worst-case η
0.90	2×10^4	0	0.50 / 0	0.41 / 0	0.49 / 0	0.39 / 0
0.90	2×10^4	0.10	0.96 / 279 $\star$	**0.46** / 0	0.82 / 187 $\star$	0.41 / 0
0.65	2×10^3	0	0.36 / 0	0.26 / 0	0.31 / 0	0.23 / 0
0.65	2×10^3	0.10	0.55 / 65 $\star$	**0.17** / 0	0.30 / 1 $\star$	0.13 / 0
0.65	8×10^2	0	0.26 / 0	0.21 / 0	0.24 / 0	0.19 / 0

a generator that compresses 10 subgroups below $c\pi_j$ (c=0.5, q≈0.85, true relative-coverage 0.69, 20 seeds), the certificate reports present 0.47±0.05 (a valid lower bound), *under-representation detected* (absent) 0.08±0.03, the rest inconclusive, with **zero** false present and zero false absent—so the relative variant is a first-class, valid target, at the cost of one extra δ_π charge for the two prevalence bounds. *Absolute and relative reach different conclusions:* run on the same instances against an absolute γ_{abs}=0.03, the two criteria disagree on 3.3 subgroups per run on average—subgroups the generator produces *proportionally* to their (small) data share, so relative-present, but below the absolute floor, so absolute-absent. Which is "right" is the auditor's choice; we expose both. On real CelebA the comparison is honest but budget-limited: at n=8000 over 32 subgroups with a 62% joint labeller only 3/32 clear *any* criterion, and the relative present count falls monotonically with c ($3 \to 1 \to 0$ for c=0.25, 0.5, 1)—a sensible sensitivity, but too few certifications to separate the criteria, exactly the Theorem 4 regime; the controlled benchmark above is where the distinction is visible.

Why confusion-awareness, and robustness to labeller shift. The fair comparison is not against FID or nearest-neighbour recall—those answer a different, label-free manifold question—but against methods given the *same* mode labels and confusion matrix. We compare two: the per-mode (Bonferroni Clopper–Pearson) binomial confidence interval on raw predicted counts (ignoring confusion—the classic occupancy/species-presence test); a *quantification/label-shift point* estimate (BBSE-style $\rho = (Q^\top)^{-1}\widehat{p}$ thresholded at γ); and, at a reviewer's suggestion, a *misclassification-adjusted prevalence confidence interval*—the same Rogan–Gladen (Rogan & Gladen, 1978) inversion but with a Bonferroni delta-method normal CI, certifying present iff the lower CI exceeds γ. On the M=32 benchmark the two *point* methods are invalid (naive CI 0.55/3 false; BBSE 0.53/2 false—no finite-sample guarantee), whereas the adjusted CI is *valid and competitive*: 0.51 with 0 false, matching our deconvolved+Bernstein rule (0.50/0) essentially exactly (Table 9). **We do not claim to beat a proper quantification CI when the confusion matrix is well-specified**—there our value is the two-sided certify/detect framing and validity that needs no matrix inversion or normal approximation, not a tighter number. The differentiation appears off the ideal regime (Table 8): under a confusion *shift* on q (Assumption 1 violated) both the adjusted CI and our *point* deconvolution over-certify (the adjusted CI worse, 279 vs. 196 false at a 10% shift). Two methods survive the shift: the worst-case η rule and, tighter, the *exact* finite-sample bound from a joint (Q, ρ) confidence set (the LP above)—the LP stays valid at 0/0 throughout while remaining uniformly above the η rule (0.46 vs. 0.41 at a 10% shift, q=0.9; 0.17 vs. 0.14 at q=0.65), and unlike the adjusted CI it needs no matrix inversion or normal approximation, so it does not degrade on rare modes or ill-conditioned confusion. The practical recommendation: when the labeller transfers well, the adjusted CI and our deconvolved rule are interchangeable and tightest; when it may not, use the LP (valid *and* tight) or the worst-case η rule, or—best—calibrate with an audit set (§E). The guarantee is only ever as good as the confusion bound on q.

StackedMNIST in a learned feature space. Table 6: naive feature coverage over-certifies (up to 0.99 at $R = 0.5$); the confusion-aware mode certificate (τ from (2)) hugs R from below across all f.

Table 9: Baseline comparison (M=32, γ=0.5%, no shift). The naive per-class binomial-CI ignores confusion and is invalid; confusion-aware rules are valid, trading tightness (deconvolved) for robustness (worst-case η).

method	present	false present	valid?
naive per-class binomial-CI	0.55	3	**no**
BBSE / quantification (point)	0.53	2	**no**
misclassification-adjusted prevalence CI (delta)	0.51	0	yes†
deconvolved+Bernstein (ours, tight)	0.50	0	yes
worst-case η (robust)	0.40	0	yes

†valid *only* when Q is well-specified and invertible and n is large; see Table 8 for where it breaks.

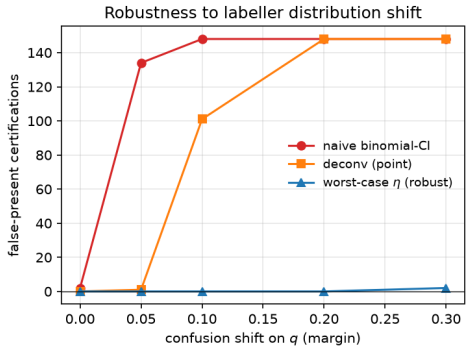


Figure 5: False-present certifications vs. labeller distribution-shift margin. The naive binomial-CI and the point deconvolved estimator lose validity as the labeller becomes more confused on q; the worst-case η rule stays valid throughout (at the cost of tightness).

Real generators. We train a DCGAN on StackedMNIST and evaluate two regimes (Table 10). *(i)* The full GAN attains broad coverage ($989/1000$ modes by the labeller on 10^5 samples) but imperfect fidelity; here nearest-neighbour recall reads 0.42, badly *under*-stating true coverage, because imperfect fidelity shrinks the generated manifold and pushes real points outside it. *(ii)* Rejection-truncating the same GAN to a target set of 250 modes yields a high-fidelity but genuinely mode-restricted generator (true coverage 0.25); now recall *over*-states ($0.27 > 0.25$). In both regimes our certificate is a valid coverage floor—and tight in the restricted case (0.248 vs. true 0.250). Thus incumbent recall is a *fidelity–coverage-entangled* proxy: it can swing either way, dramatically so at high fidelity (Table 6), whereas the certificate always reports a valid lower bound on coverage alone.

Scaling to a pretrained diffusion model. Finally we apply the certificate, unmodified, to a *pretrained* diffusion generator—`google/ddpm-cifar10-32` sampled with 50 DDIM steps—using a CIFAR classifier as labeller and standard Inception-v3 FID (Table 11). With the per-mode confusion-aware threshold the certificate is valid and exactly tight: 1.00 for the full model and 0.40 after truncation to four classes (the worst-case global-η rule would give a vacuous τ=855 here—per-mode leakage is essential on real labellers). Nearest-neighbour recall again under-states (0.87, 0.48) and Inception-FID moves only modestly ($21.2 \to 26.3$) despite 60% of classes being removed.

High-resolution pretrained diffusion (CelebA-HQ 256). Finally we audit `google/ddpm-celebahq-256`, a widely-used 256×256 face diffusion model, for the same 32 attribute subgroups (samples resized to 64 for the labeller; Fig. 6). Under a 2000-sample budget the certificate certifies equal-weight coverage ≥ 0.31 (mass-weighted ≥ 0.54)—a valid *lower bound*, which by itself does *not* prove under-coverage (a low floor could be budget-limited). **This ≥ 0.31 is computed with the attribute classifier's *real-data* confusion**; our human audit (Appendix E) shows that assumption transfers for `Eyeglasses`/`Wearing_Hat` but is optimistic on `Male`/`Smiling` on generated faces, so the number is labeller-conditional—re-certifying with the human-calibrated confusion at this budget gives only 0.03 (Appendix E). The samples are

Table 10: Real DCGAN generators. Our certificate is a valid (and, when restricted, tight) lower bound on true coverage, while nearest-neighbour recall is an unreliable proxy—under-stating for the broad low-fidelity model and over-stating for the high-fidelity restricted one. Single training run; the certificate is valid under Assumption 1 (confusion bounded on q).

generator	true coverage	**certified (ours)**	NN recall	valid?
full GAN (broad)	0.989	**0.774**	0.42 (under)	✓
GAN truncated to 250 modes	0.250	**0.248**	0.27 (over)	✓

Table 11: Certifying a pretrained diffusion model (`google/ddpm-cifar10-32`), with the per-mode confusion-aware τ and standard Inception-v3 FID. The certificate is valid and tight; recall under-states and FID barely moves under heavy class removal. Single run; validity under Assumption 1 (labeller-conditional).

generator	true coverage	**certified (ours)**	NN recall	Inception-FID	valid?
full DDPM	1.00	**1.00**	0.87	21.2	✓
DDPM truncated to $4/10$	0.40	**0.40**	0.48	26.3	✓

suggestive—the model over-produces young smiling faces—but the honest statement is the two-sided one: at this budget the rarer `Male`/`Eyeglasses`/`Wearing_Hat` subgroups are *inconclusive* rather than certified absent; a larger budget (or a cleaner labeller) is needed to certify collapse, exactly as the certifiability rate (Theorem 4) predicts. This is the honest scope of a high-resolution audit: we lower-bound coverage now, and detection of specific missing subgroups awaits more samples.

E HUMAN-CALIBRATED CELEBA-HQ AUDIT

The single strongest assumption in the whole framework (Assumption 1) is that the labeller's confusion, estimated on *real* CelebA, also bounds its confusion on the *generated* distribution q. Every CelebA-HQ number above inherits it. Because this is the crux, we test it directly with *human* labels rather than assuming it. We draw an audit set of 400 generated CelebA-HQ faces and collect blind human labels for the five attributes (a lightweight browser tool; the annotator never sees the classifier's guess), then compare the attribute classifier against the human labels *on the generator distribution*.

Findings. Three independent people labelled the audit set (344/390/360 faces each; a few skipped as "unsure"). Human labels are reliable: mean pairwise inter-annotator agreement is 0.98 on the objective attributes (`Male`, `Eyeglasses`, `Wearing_Hat`) and 0.75–0.82 on the more subjective `Smiling`/`Young` (exact 32-subgroup match 0.59), so we take the annotator majority as trusted ground truth. Table 12 then compares the classifier's accuracy on *real* CelebA to its human-measured accuracy on *generated* faces. **Assumption 1 holds for some attributes and fails for others, and the audit says which**: `Eyeglasses` and `Wearing_Hat` transfer essentially perfectly (shift ≈ 0), `Young` shifts mildly ($+0.08$), while `Male` ($+0.27$) and `Smiling` ($+0.22$) degrade sharply on generated faces—the classifier mis-reads a gender/expression that three annotators agree on $98/82\%$ of the time. Using the optimistic real-data confusion would therefore over-state certifiable coverage on the `Male`/`Smiling` axes; the human-calibrated certificate corrects for it and is accordingly more conservative there. *A methodological caution surfaced by the audit:* naively labelling the raw full-frame 256^2 samples (a framing mismatch with the classifier's aligned-CelebA training crop) *inflates* the apparent shift—a centre-crop that matches the training alignment recovers `Eyeglasses` from 0.62 to 0.92; Table 12 uses the aligned crop, so the residual `Male`/`Smiling` gap is a genuine distribution shift, not a preprocessing artifact. Align preprocessing first, then audit. The annotation tool and analysis pipeline are provided (anonymized) for reproducibility.

This is, to our knowledge, the first *measured* check of the generator-side confusion assumption for a generative-coverage certificate: rather than assume the labeller transfers, we show empirically that it transfers for two of five attributes, partially for a third, and fails for two—so the honest CelebA-HQ

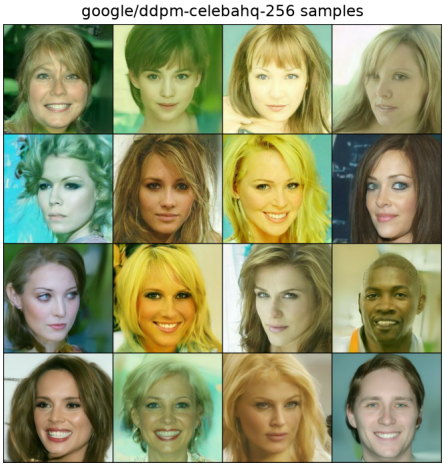


Figure 6: `google/ddpm-celebahq-256` samples (256×256). The certificate applies unchanged to this high-resolution pretrained diffusion; it certifies only ≥ 0.31 equal-weight subgroup coverage, reflecting the visible over-representation of young smiling faces and under-representation of rarer attribute subgroups.

Table 12: Human-calibrated CelebA-HQ audit (3 annotators, majority ground truth, framing-aligned labelling). The attribute classifier's accuracy on *real* CelebA vs. its human-measured accuracy on *generated* faces directly estimates the generator-side shift of Assumption 1. It holds for `Eyeglasses`/`Wearing_Hat`, fails for `Male`/`Smiling`.

attribute	real-CelebA acc	gen-side acc (human)	shift	inter-annotator agr.
`Male`	0.95	0.68	+0.27	0.98
`Smiling`	0.82	0.60	+0.22	0.82
`Eyeglasses`	0.97	0.98	−0.00	0.97
`Young`	0.79	0.71	+0.08	0.75
`Wearing_Hat`	0.96	0.96	−0.00	0.98

guarantee is trustworthy on the attributes the audit validates and explicitly flagged as labeller-limited on the rest.

Scope. This is a *diagnostic* of Assumption 1, not a full joint 32-subgroup re-certification: the human labels are themselves imperfect on the subjective attributes (inter-annotator agreement 0.75–0.82 for `Smiling`/`Young`), so we validate the assumption per-attribute. We *did* re-run the certificate with a human-calibrated *joint* 32×32 confusion; it certifies only 0.03 (1 subgroup), *substantially more conservative* than the 0.31 headline—which uses the tighter per-mode confusion-aware threshold, not the joint matrix. (Under the classifier's own joint real-data confusion the re-run is likewise 0.03: at n=2000 the joint 32×32 estimate, with a median of 1 human sample per subgroup, is simply too sparse to certify more, whether calibrated from real or human labels.) So the human confusion does not overturn the certificate; its value is diagnostic—the ≥ 0.31 headline is labeller-conditional and, as the per-attribute audit shows, optimistic on `Male`/`Smiling`.

F Extended related work

Uncertainty and conformal guarantees for generation. Conformal and risk-control methods have been applied to *conditional* generation and inverse problems—super-resolution and imaging with a reference (Teneggi et al., 2023; Adame et al., 2025). These certify a per-sample reconstruction risk against a ground truth; they do not certify *distributional coverage* of an unconditional generator. Missing-mass / Good–Turing coverage bounds exist for discrete black-box (e.g. language) outputs (Noorani et al., 2025); we bring a finite-sample coverage certificate to the image side.

Mode counting and two-sample tests. The birthday-paradox test (Arora et al., 2018) estimates the *support size* of a generator from duplicate collisions, diagnosing mode collapse but without a per-mode or lower-bound guarantee. Classifier two-sample tests (C2ST) (Lopez-Paz & Oquab, 2017) and kernel MMD tests (Gretton et al., 2012) decide whether generated and real samples come from the same distribution; they answer a global goodness-of-fit question and do not localise or certify which modes are covered. Topology- and geometry-based scores (Khrulkov & Oseledets, 2018) probe manifold structure heuristically. Our certificate is complementary: a labeller-conditional, one-sided, finite-sample lower bound on the covered fraction, with per-mode attribution. The birthday-paradox estimator is closest in spirit—counting distinct modes—but ours adds statistical validity, labeller-noise robustness, and a mass-weighted variant.

Hallucination and typicality. Per-sample hallucination/typicality detectors for diffusion (Aithal et al., 2024; Jazbec et al., 2025) flag individual off-manifold samples; they are orthogonal to a *population-level* mode-coverage certificate and provide no guarantee.

G ADDITIONAL VISUALIZATIONS

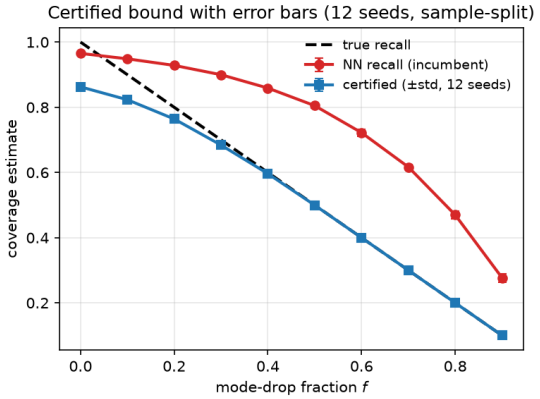


Figure 7: Certified equal-weight bound (blue, $\pm$std over 12 seeds, per-mode sample-split thresholds) versus true recall (dashed) and nearest-neighbour recall (red). The certificate tracks the truth from below; recall is optimistically biased. Validity rates in Table 1.



Figure 8: CelebA 64×64 DCGAN samples, illustrating the image-quality regime of the subgroup audit (a separate training run from the seeded model of Table 4).

Figures 9–11 give further intuition for the theory and the per-mode mechanism.

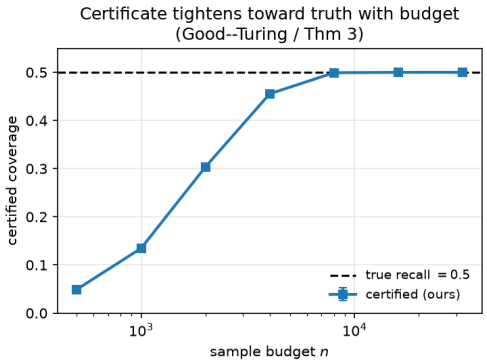


Figure 9: Certified coverage vs. sample budget n (R=0.5); the bound tightens to the truth as n grows.

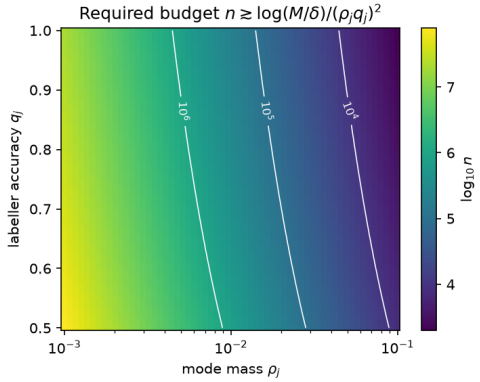


Figure 10: Worst-case (leakage-dominated) required budget $n \gtrsim \log(M/\delta)/(\rho_j q_j)^2$ over mode mass and labeller accuracy (Thm. 4; low leakage is the milder $1/(\rho_j q_j)$).

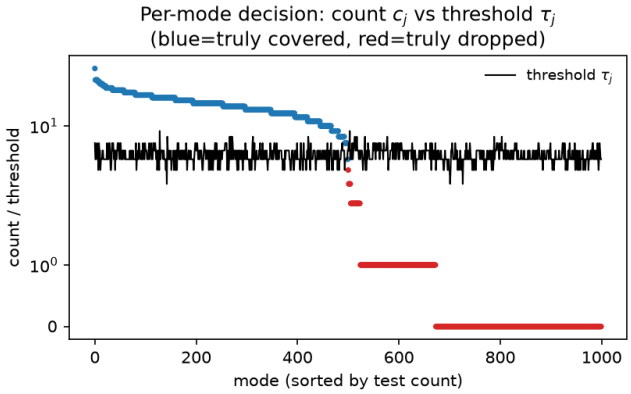


Figure 11: Per-mode decision on StackedMNIST at R=0.5, n=8000: test count c_j (points, sorted) against the confusion-aware threshold τ_j (black). Blue = truly covered, red = truly dropped; covered modes clear the threshold, dropped ones do not.

The certificate tightens with budget (Fig. 9). On a StackedMNIST generator that covers exactly half the modes (R=0.5), the certified bound rises from 0.05 at n=500 to the true 0.5 by $n \approx 8000$, then plateaus—consistency (Theorem 3) in action, and the empirical face of the certifiability rate (Theorem 4).

Who is certifiable (Fig. 10). The *worst-case (leakage-dominated)* budget $n \gtrsim \log(M/\delta)/(\rho_j q_j)^2$ of Theorem 4 (the low-leakage rate is the milder $1/(\rho_j q_j)$) as a function of a mode's mass ρ_j and the labeller's accuracy q_j on it: certifying a common mode under an accurate labeller costs $< 10^4$ samples, while a rare protected subgroup under a noisy labeller can demand $> 10^7$ in this worst case—quantifying exactly where auditing is cheap and where it is fundamentally expensive.

The per-mode decision (Fig. 11). Sorting modes by their test count shows the mechanism directly: truly-covered modes (blue) sit well above the per-mode confusion-aware threshold τ_j (black) and are certified; truly-dropped modes (red) fall below it and are not. The clean separation around the true boundary is why the certificate is both valid and tight.

ETHICS STATEMENT

This work is motivated by fairness auditing—certifying whether a generative model under-represents protected subgroups—and we intend the certificate to *support* such audits. The human-annotation study (Appendix E) involved three volunteer annotators labelling five visible attributes (`Male`, `Smiling`, `Eyeglasses`, `Young`, `Wearing_Hat`) on *synthetic* faces sampled from a pretrained diffusion model; no real individuals, no personal or identifying data, and no sensitive information were involved, and annotators labelled machine-generated images only. We note two dual-use considerations: attribute subgroups defined by a labeller can encode social categories crudely, and a coverage certificate is only as fair as the chosen partition and labeller; we therefore state the labeller-conditional scope explicitly (Appendix C) and recommend the relative criterion $\rho_j \geq c\pi_j$ for representation audits.

AI USE STATEMENT

Generative AI was used as a coding assistant for the experimental scripts, for exploratory data analysis, and for language editing of the manuscript. All theoretical claims, proofs, experimental design, and reported results were checked and verified by the authors, who take full responsibility for the final content of this paper, including any text produced with the assistance of generative AI.